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(54) **PROGRESSIVELY CHILLED GAS SEPARATION TANKS**

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F25J 3/06 (2006.01)

(52) **U.S. Cl.**
CPC **F25J 3/069** (2013.01); **F25J 3/0695** (2013.01); **F25J 2205/02** (2013.01); **F25J 2215/10** (2013.01); **F25J 2270/91** (2013.01)

(58) **Field of Classification Search**
CPC **F25J 3/069**; **F25J 3/0695**; **F25J 2215/30**; **F25J 2270/91**; **F25J 1/007**; **F25J 1/001**; **F25J 1/02**; **B01D 59/48**

See application file for complete search history.

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Primary Examiner — Frantz F Jules

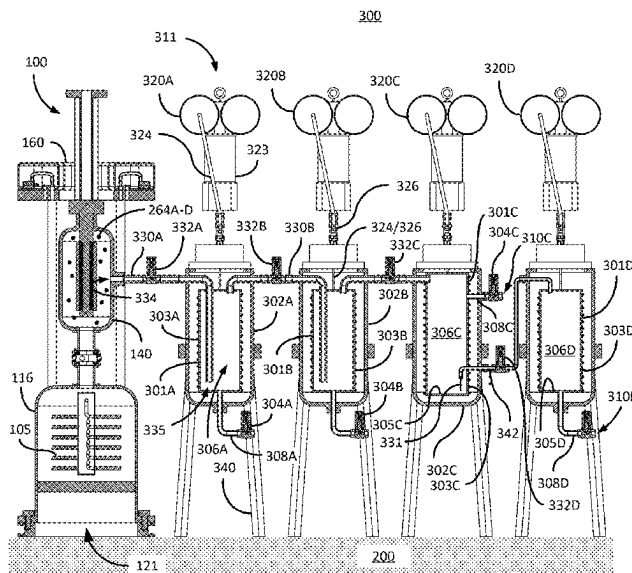
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(57) **ABSTRACT**

Disclosed is a gas separation arrangement for separating and containing various types of gas received from a collection chamber. In certain embodiments, the gas separation arrangement is envisioned to assist in mining select gases from the Moon. The arrangement includes plurality of progressively chilled separation tanks that successively separate out different species of gas. The separation tanks are connected to and receive the various gas species from the collection chamber. Each separation tank can have an independent heat exchanger that maintains the respective separation tank at a temperature that selectively liquifies/precipitates out a target species of gas allowing the remaining gas to move to a successive separation tank. Ultimately, He-3 and He-4 are separated from one another to be collected, like the other gas species.

20 Claims, 19 Drawing Sheets



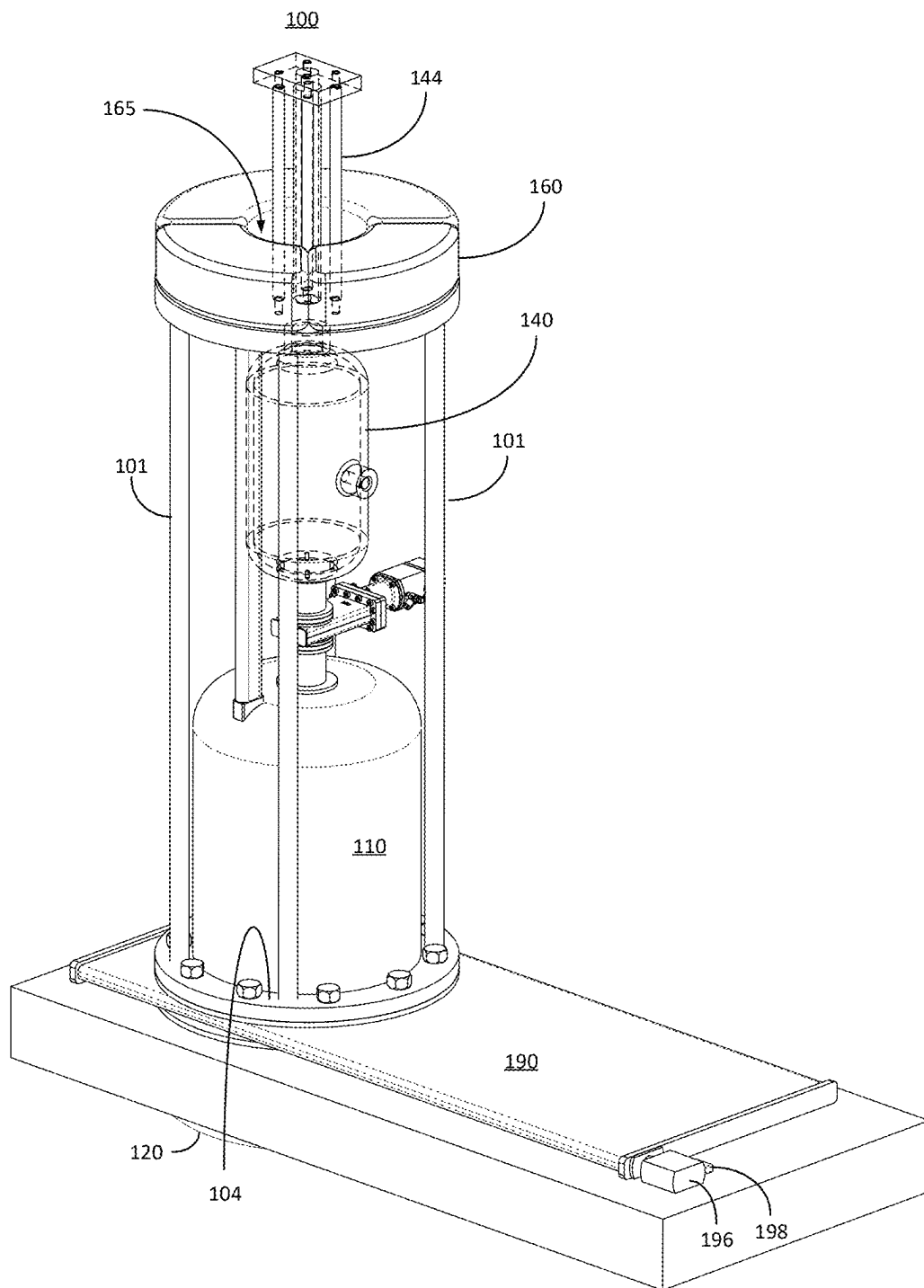
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**FIG. 1A**

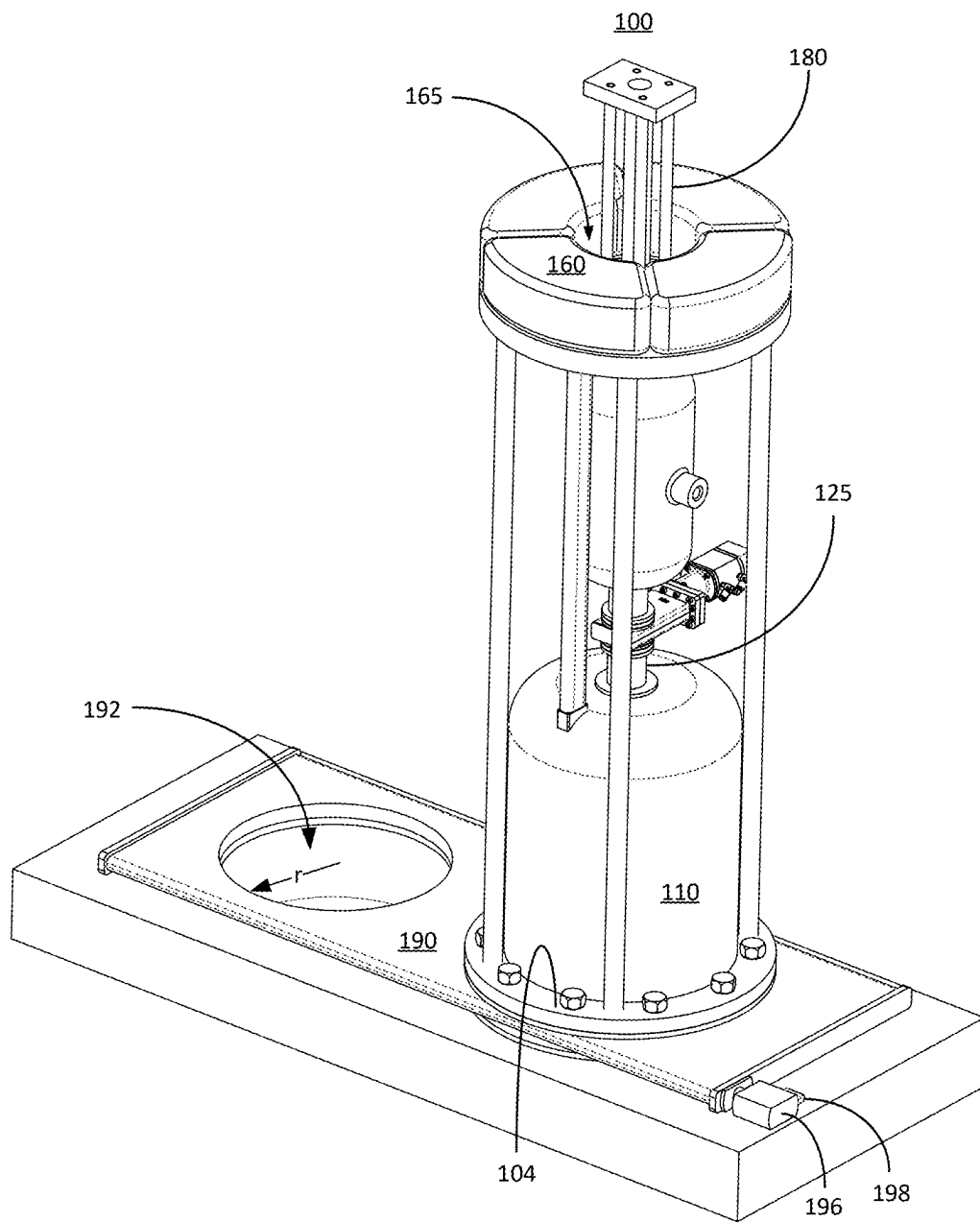
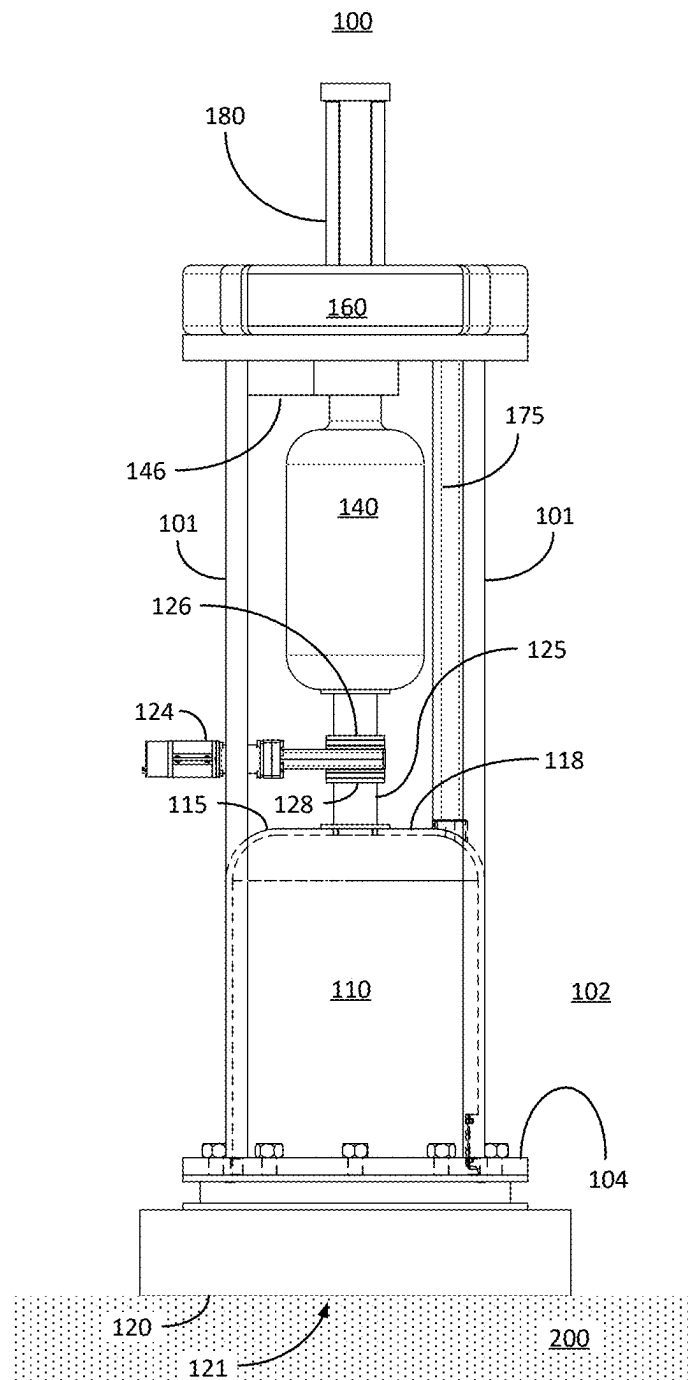


FIG. 1B

**FIG. 1C**

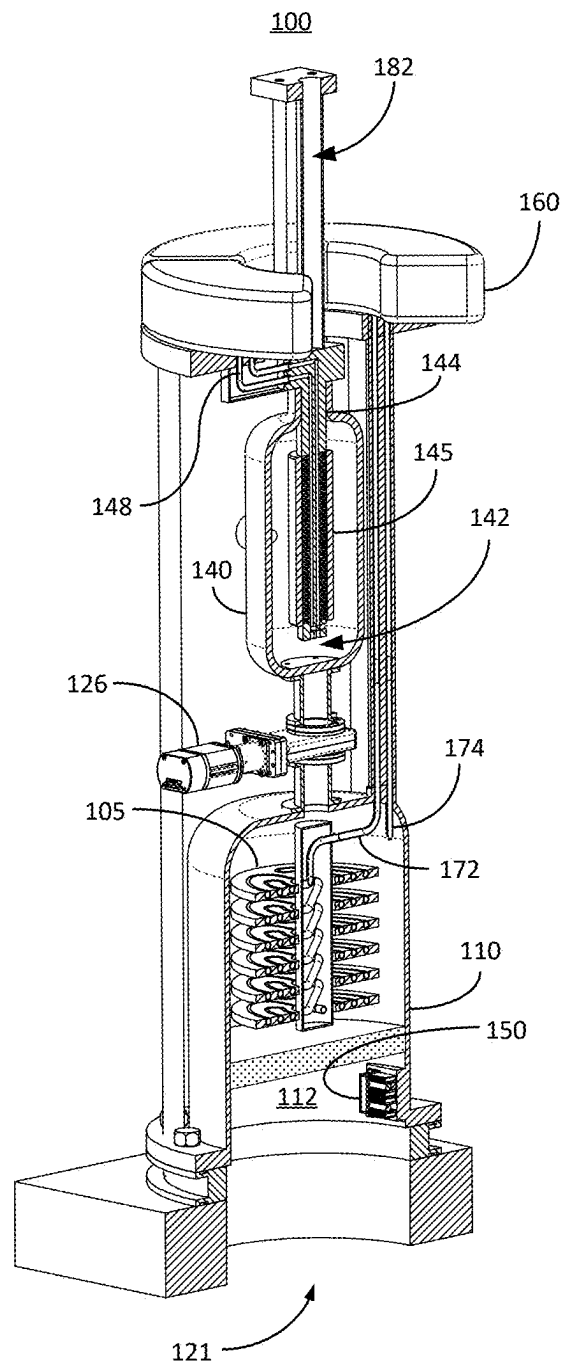


FIG. 1D

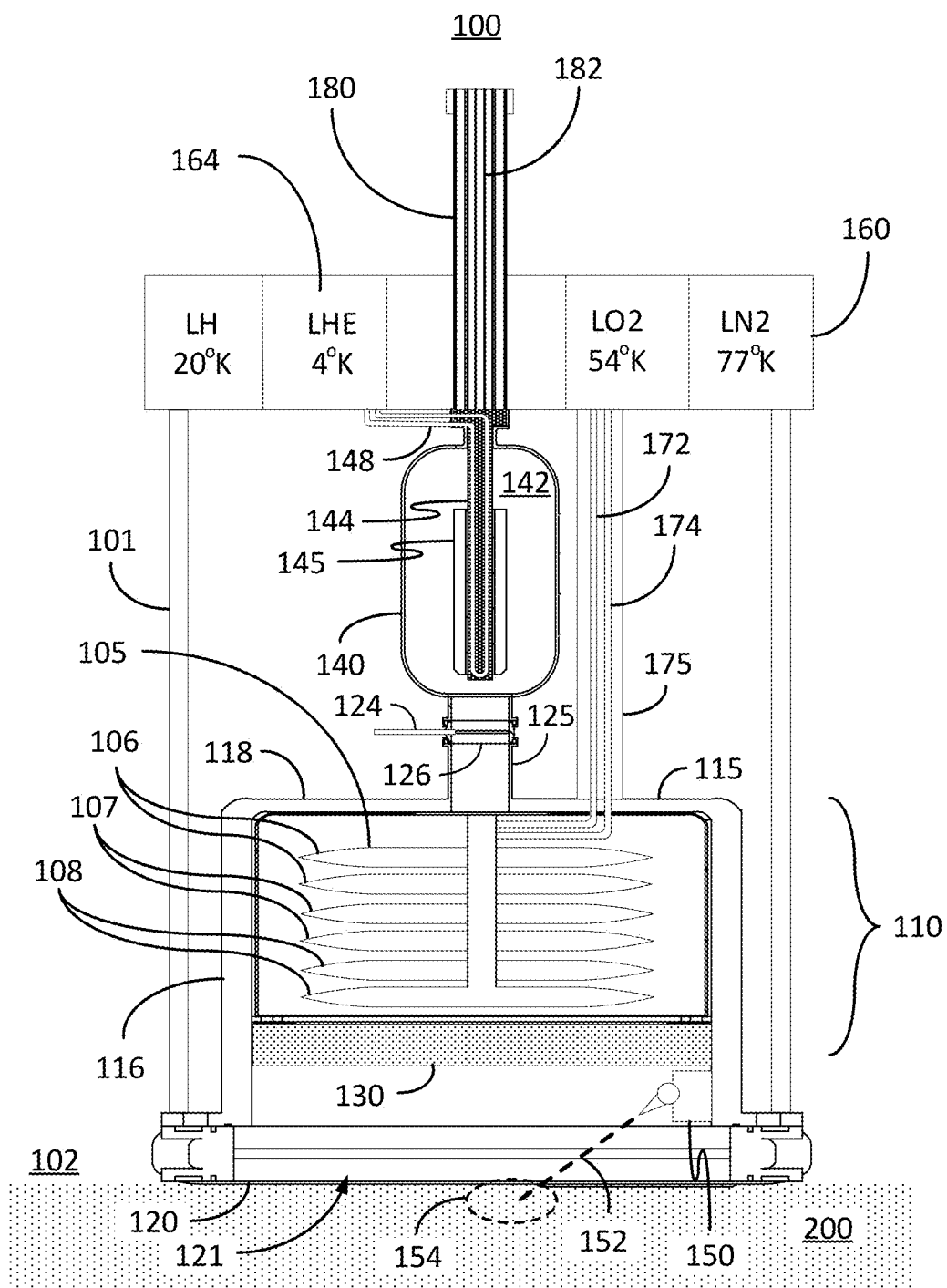


FIG. 1E

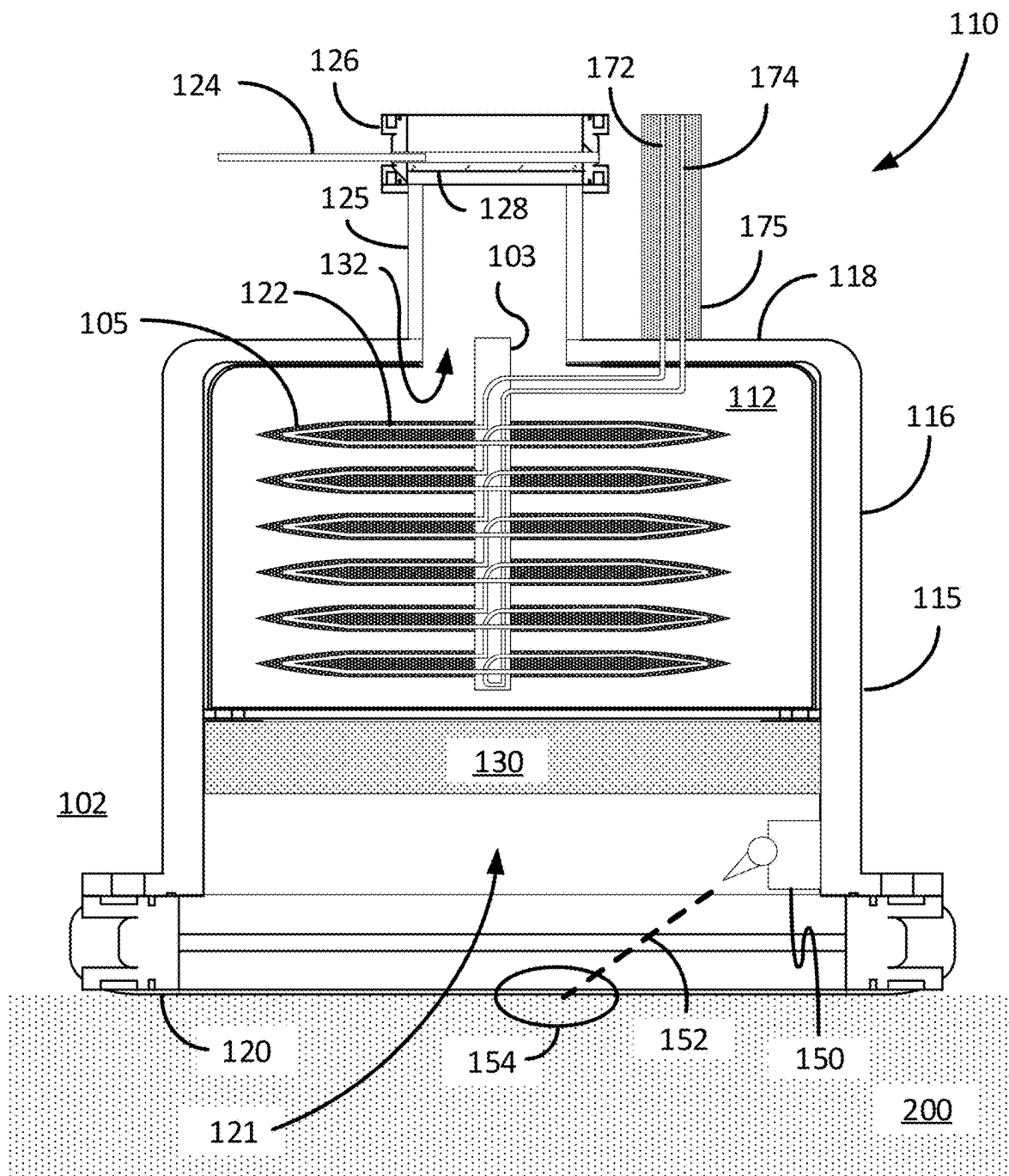
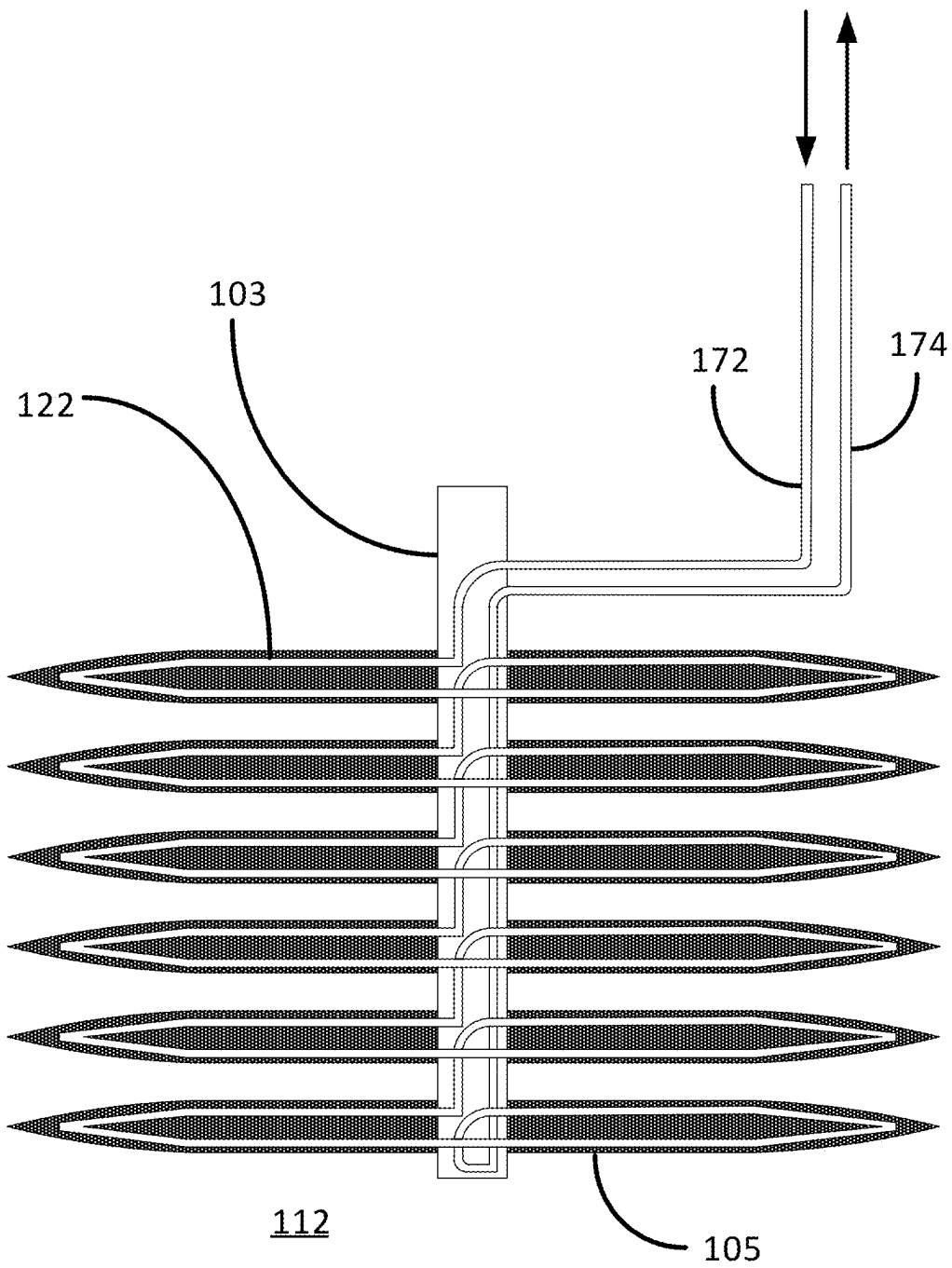


FIG. 2A

**FIG. 2B**

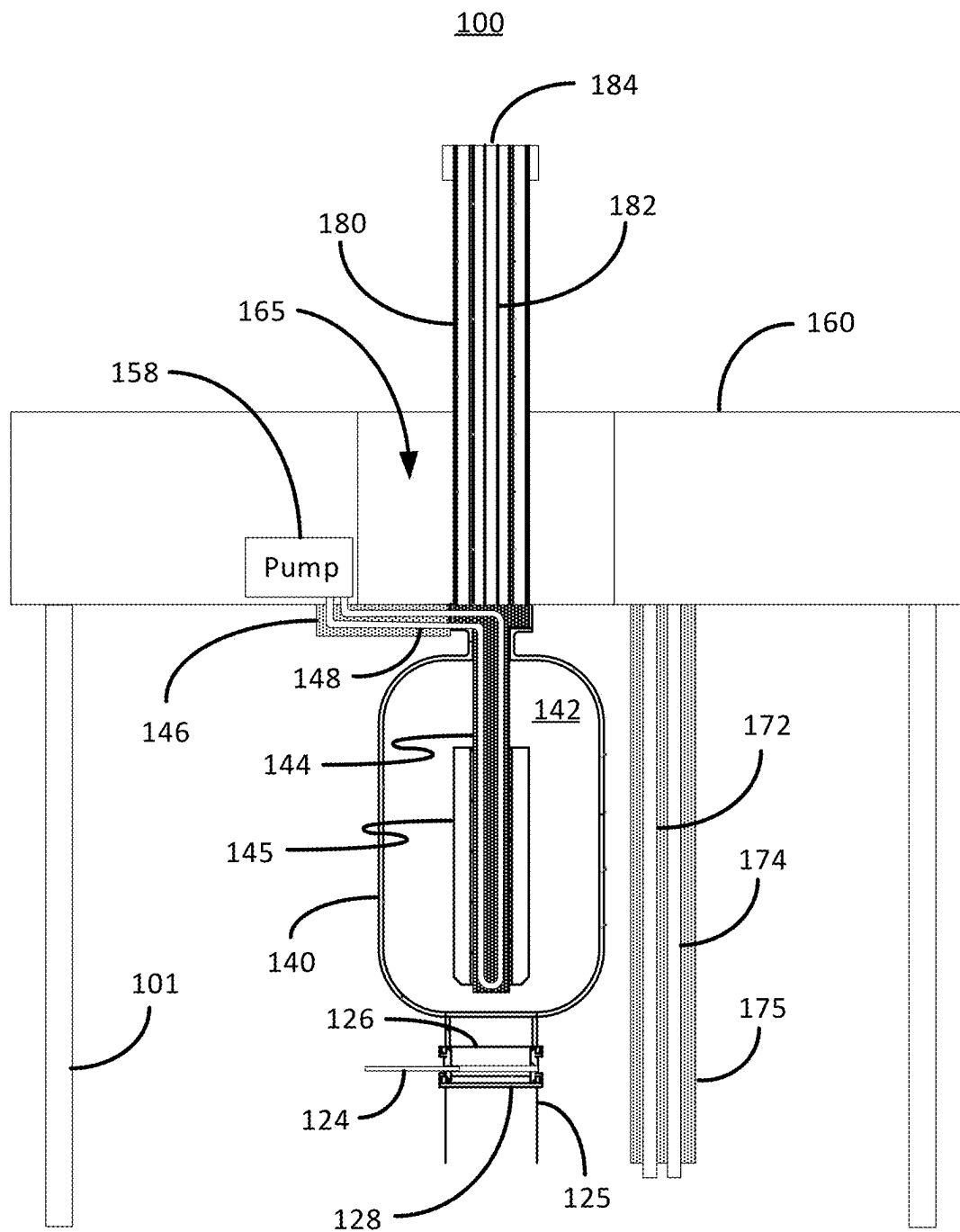
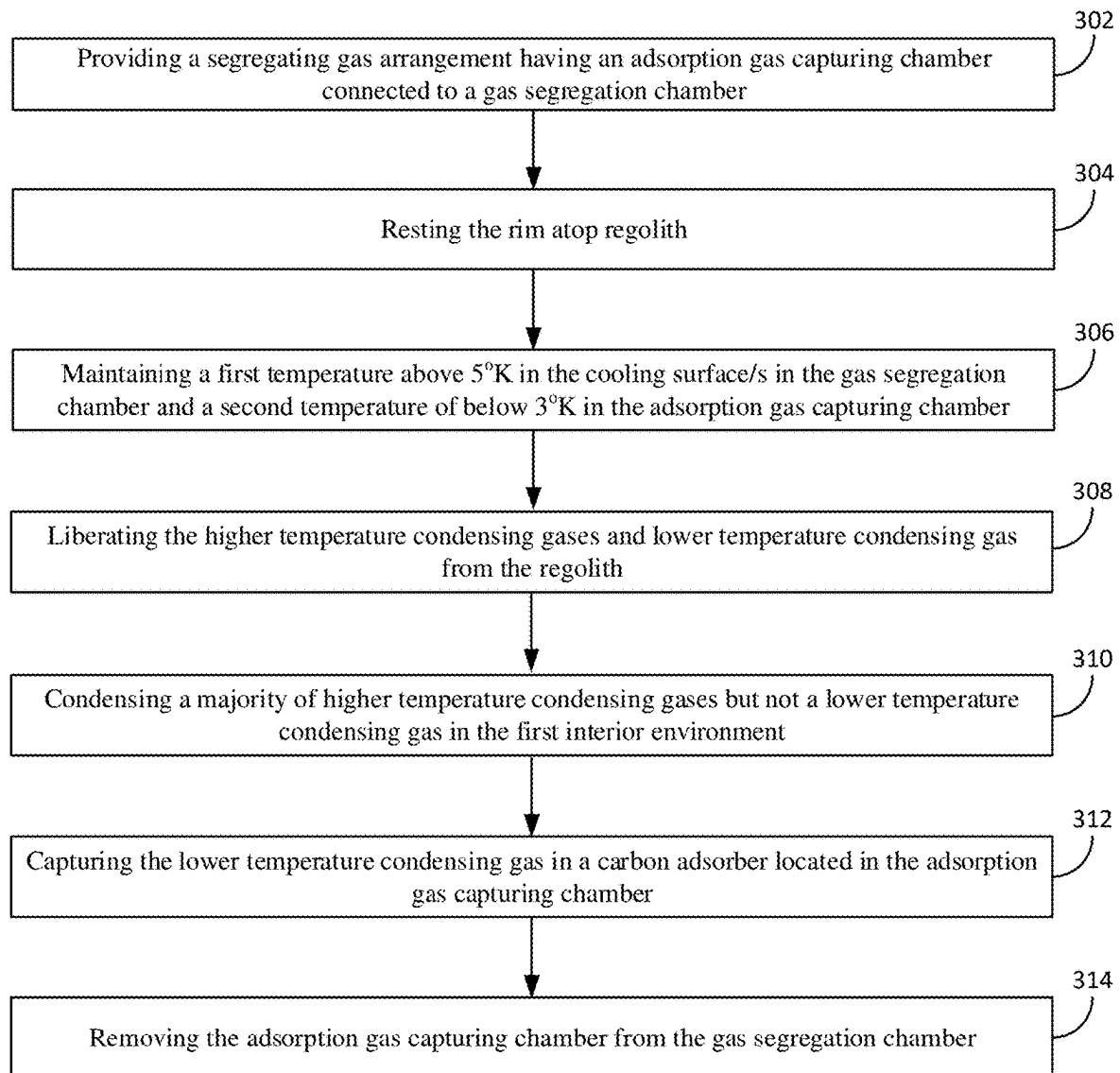


FIG. 3

**FIG. 4**

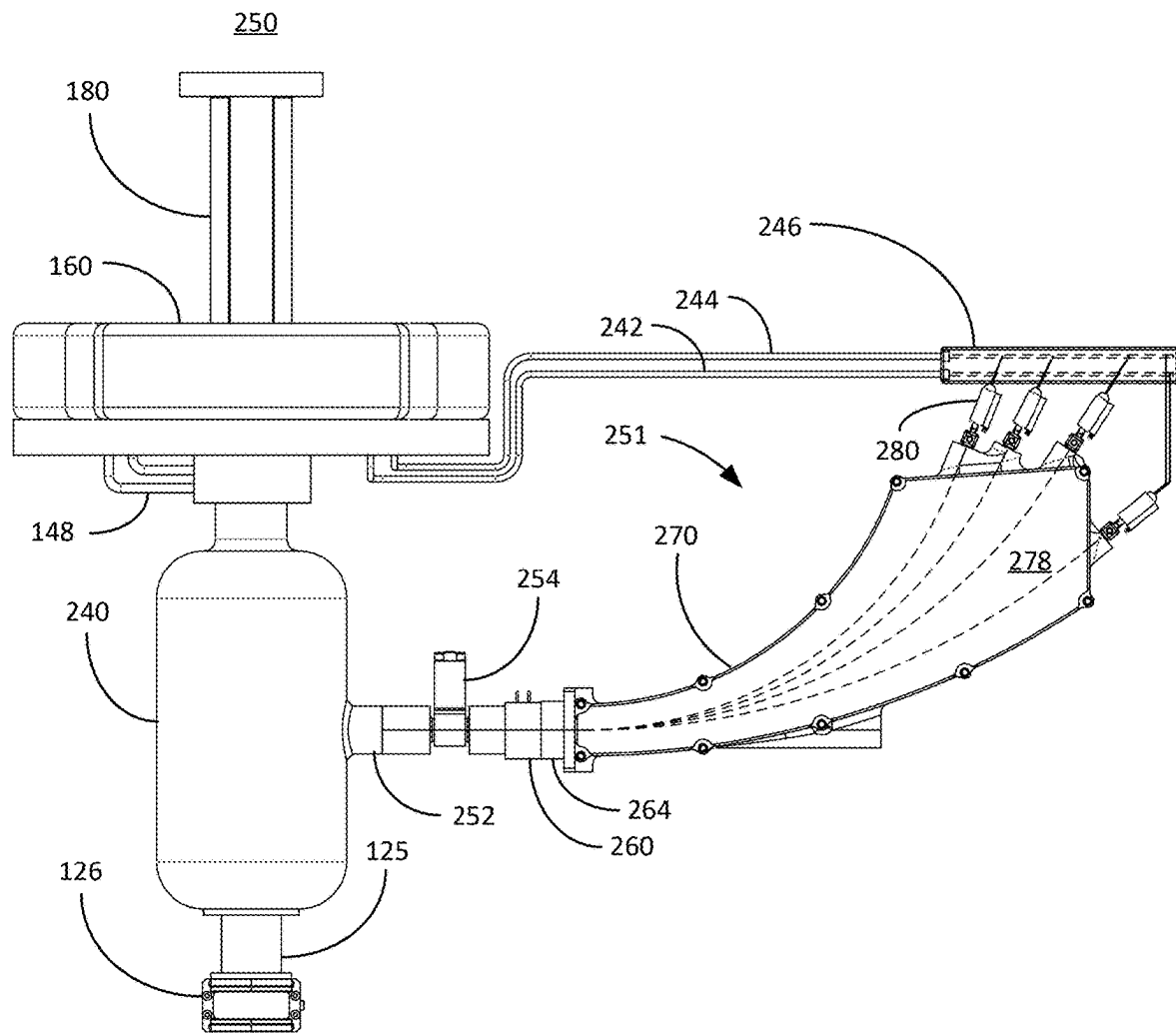


FIG. 5A

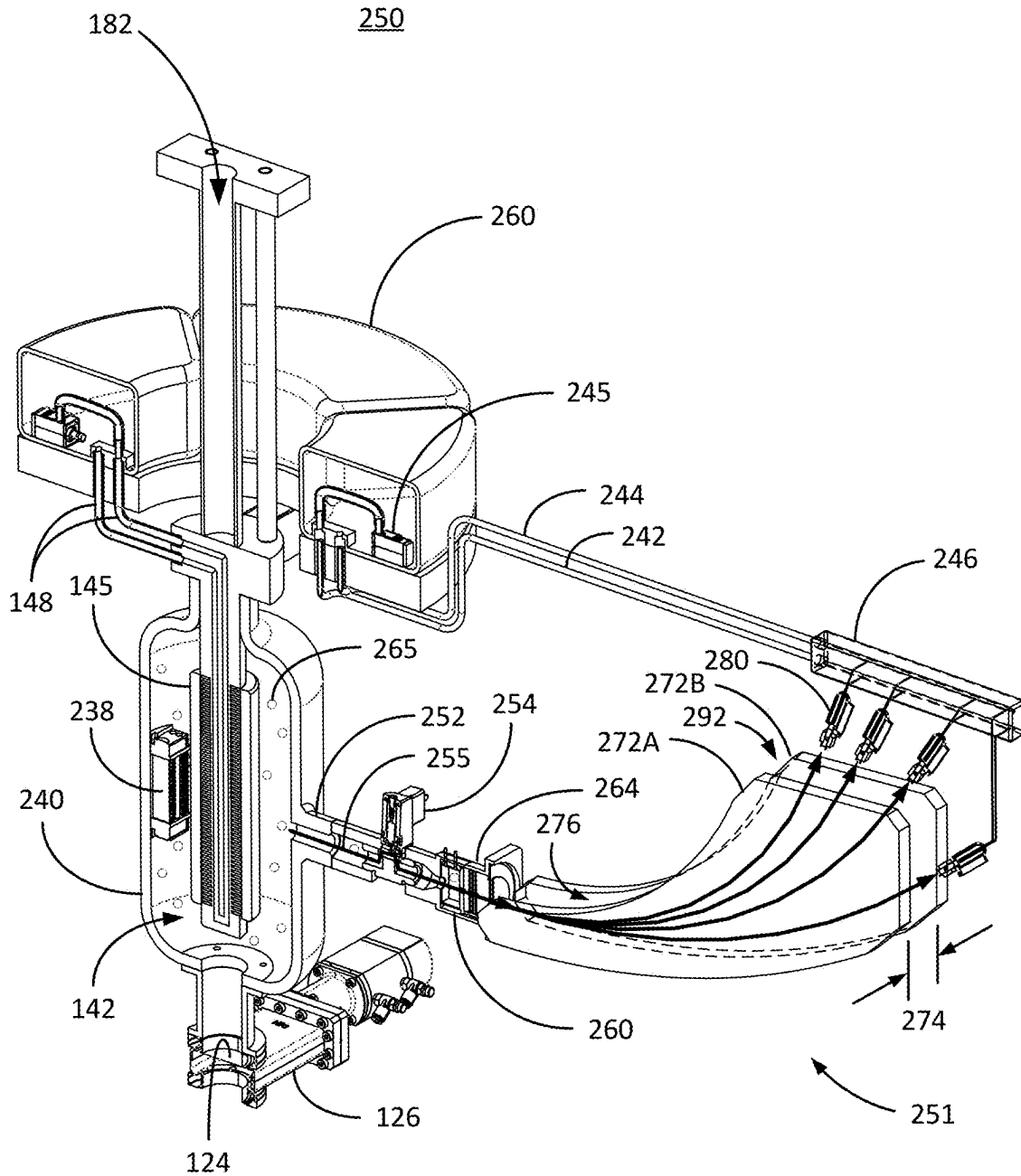


FIG. 5B

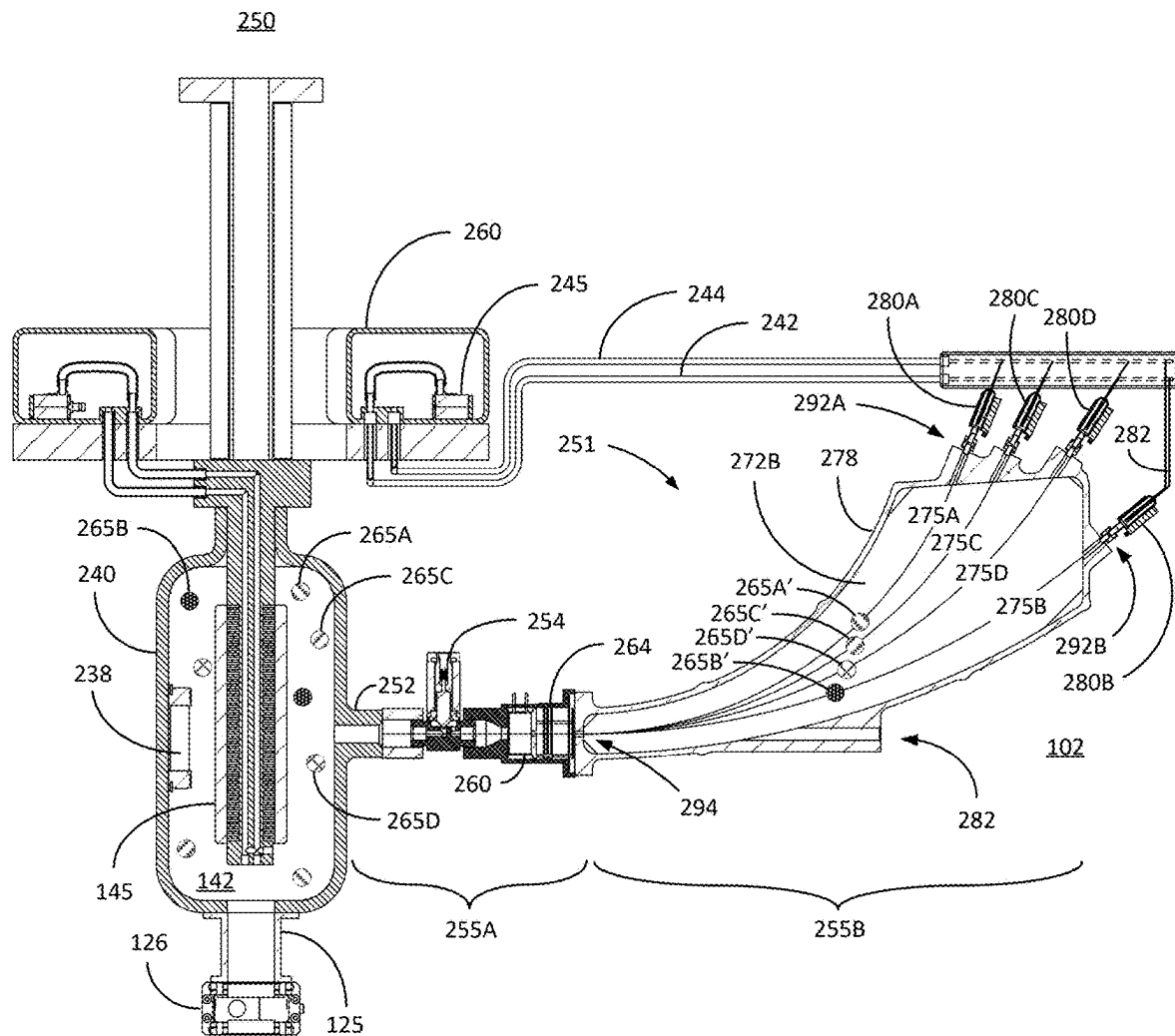


FIG. 5C

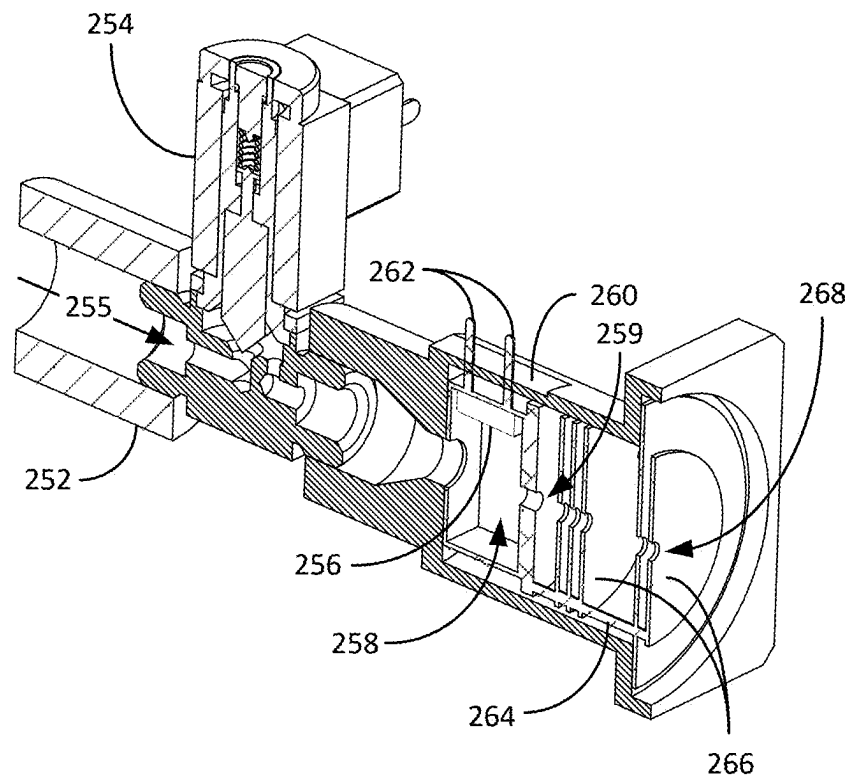


FIG. 5D

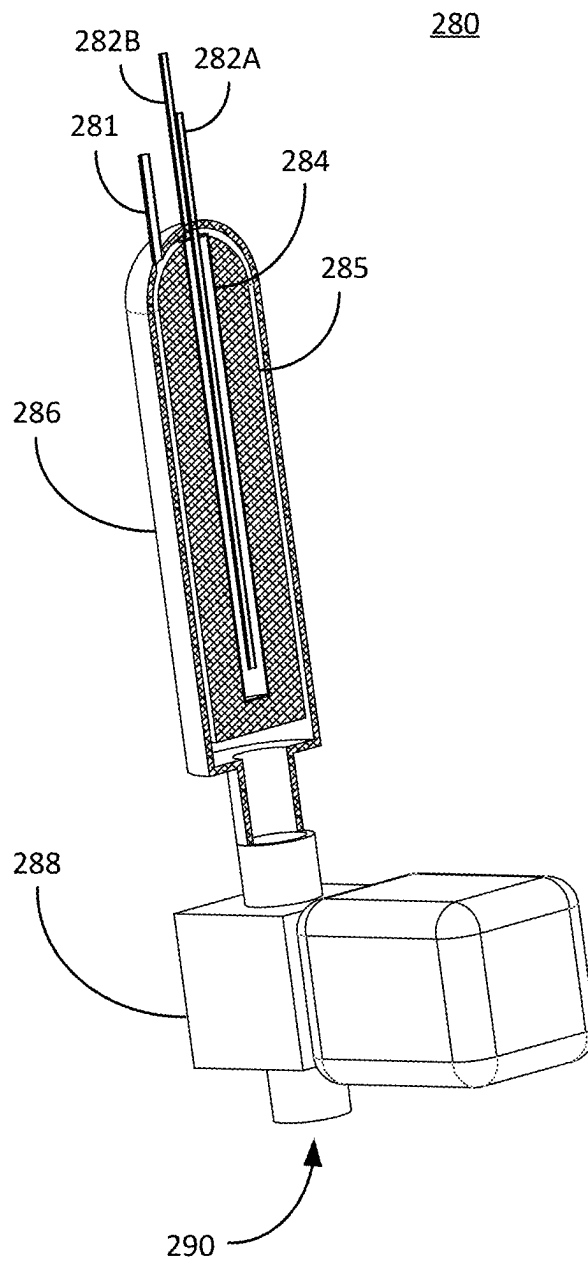


FIG. 5E

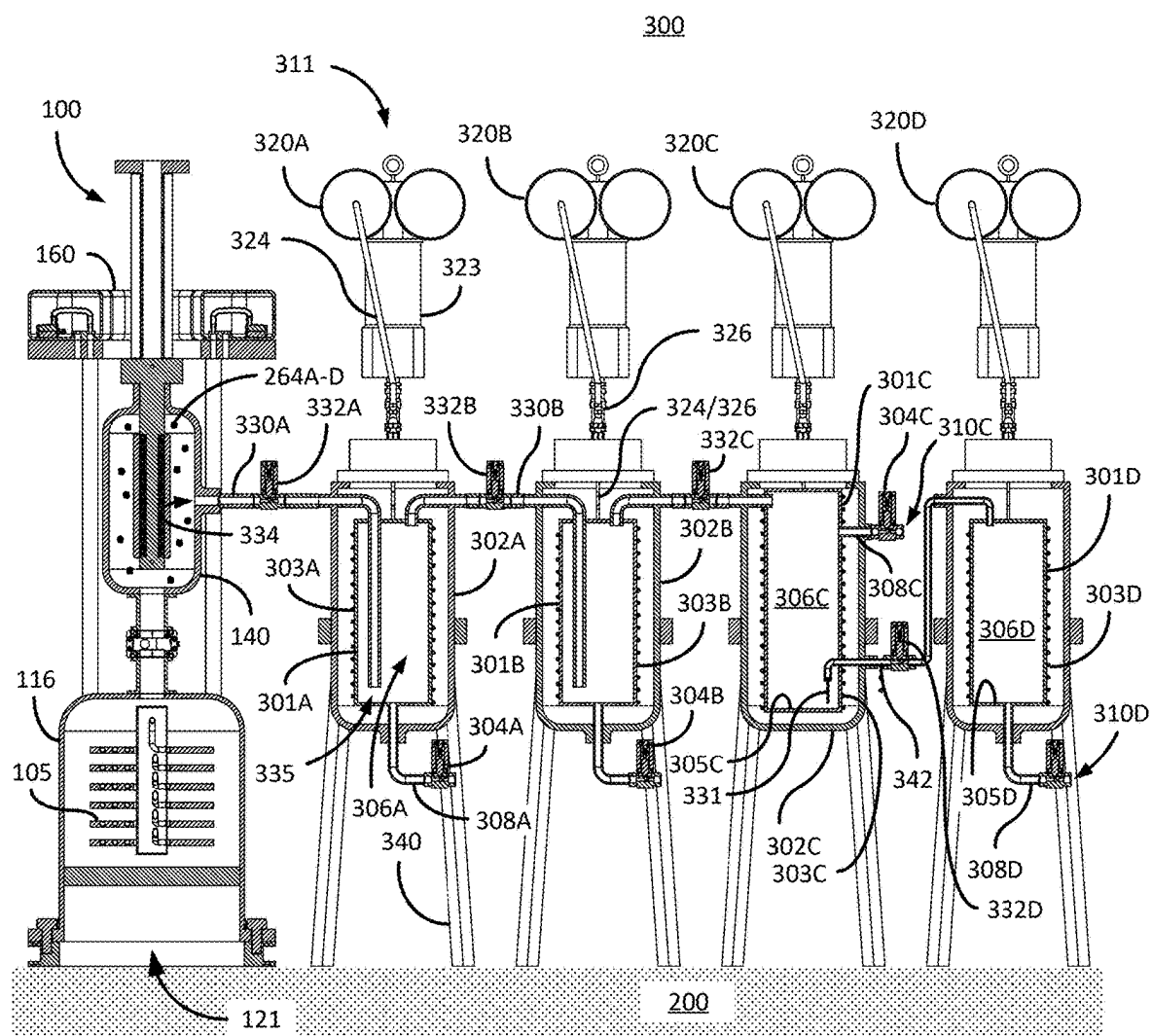


FIG. 6A

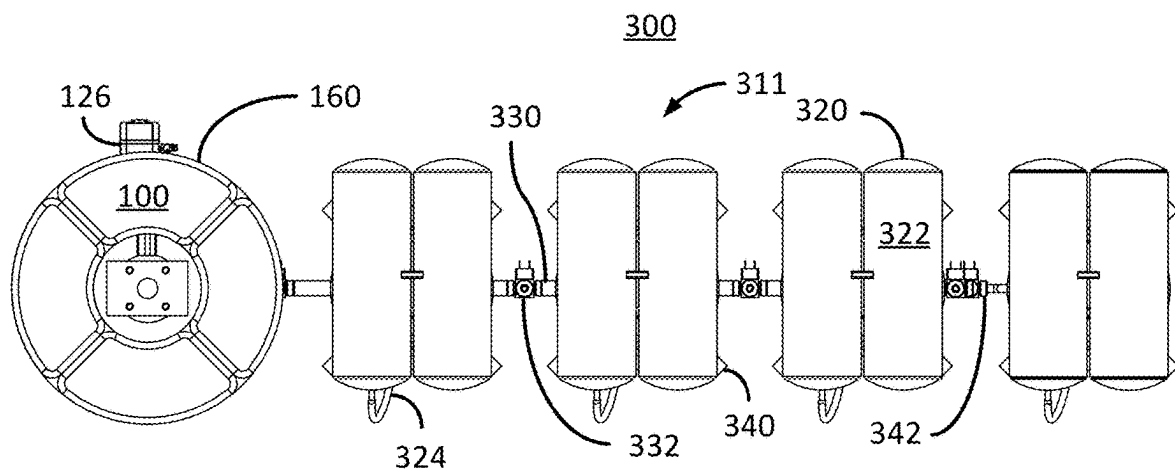


FIG. 6B

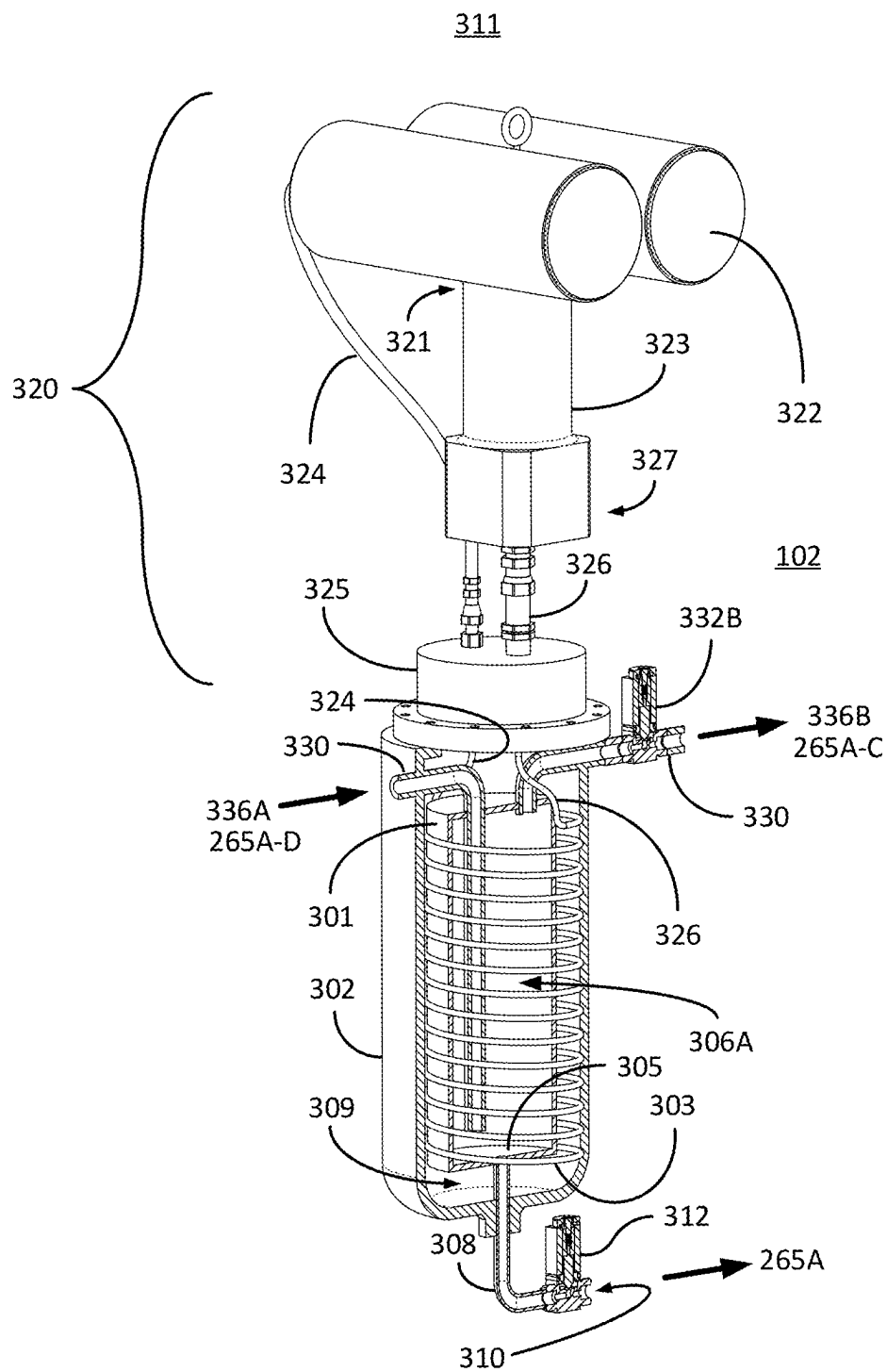


FIG. 6C

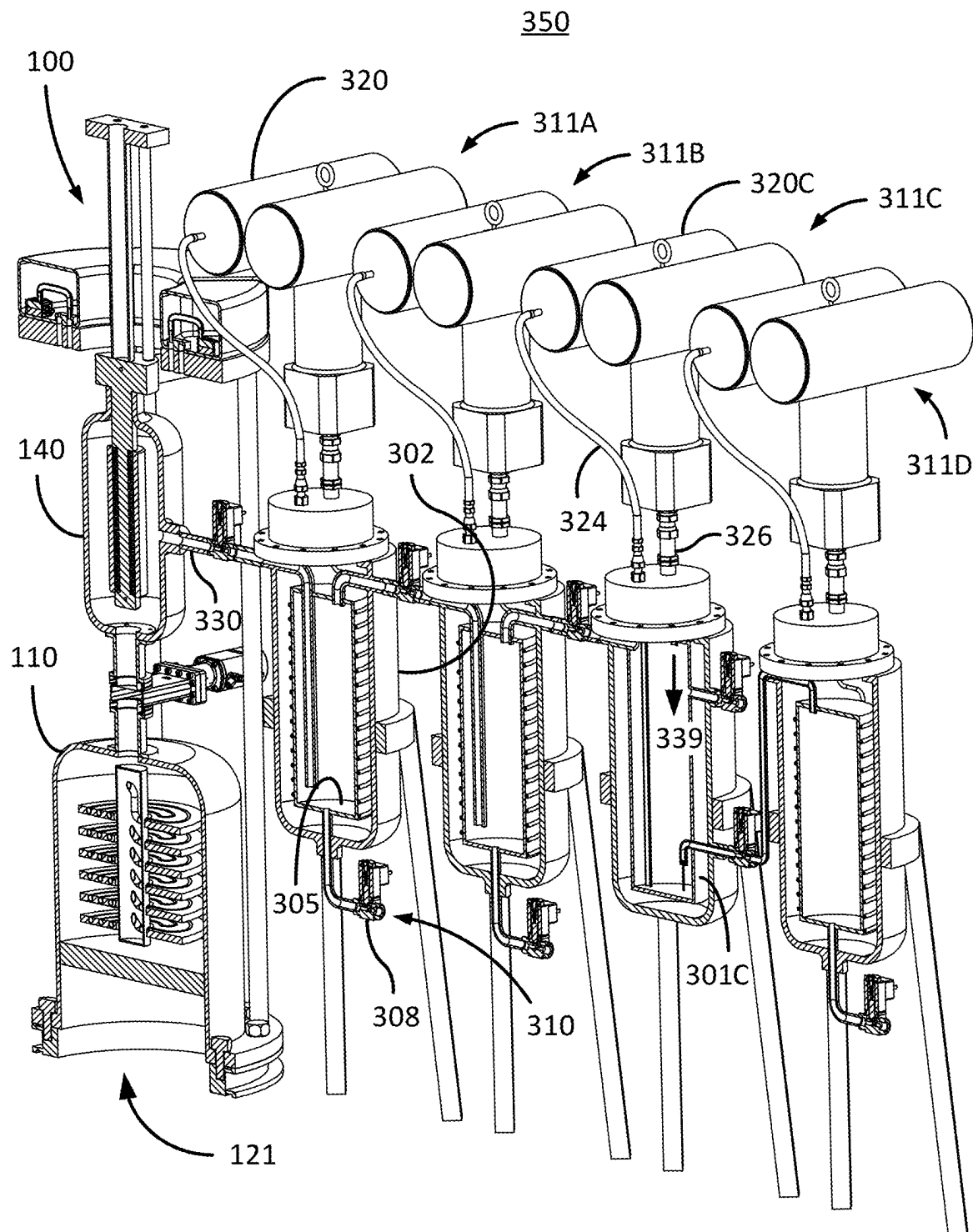


FIG. 6D

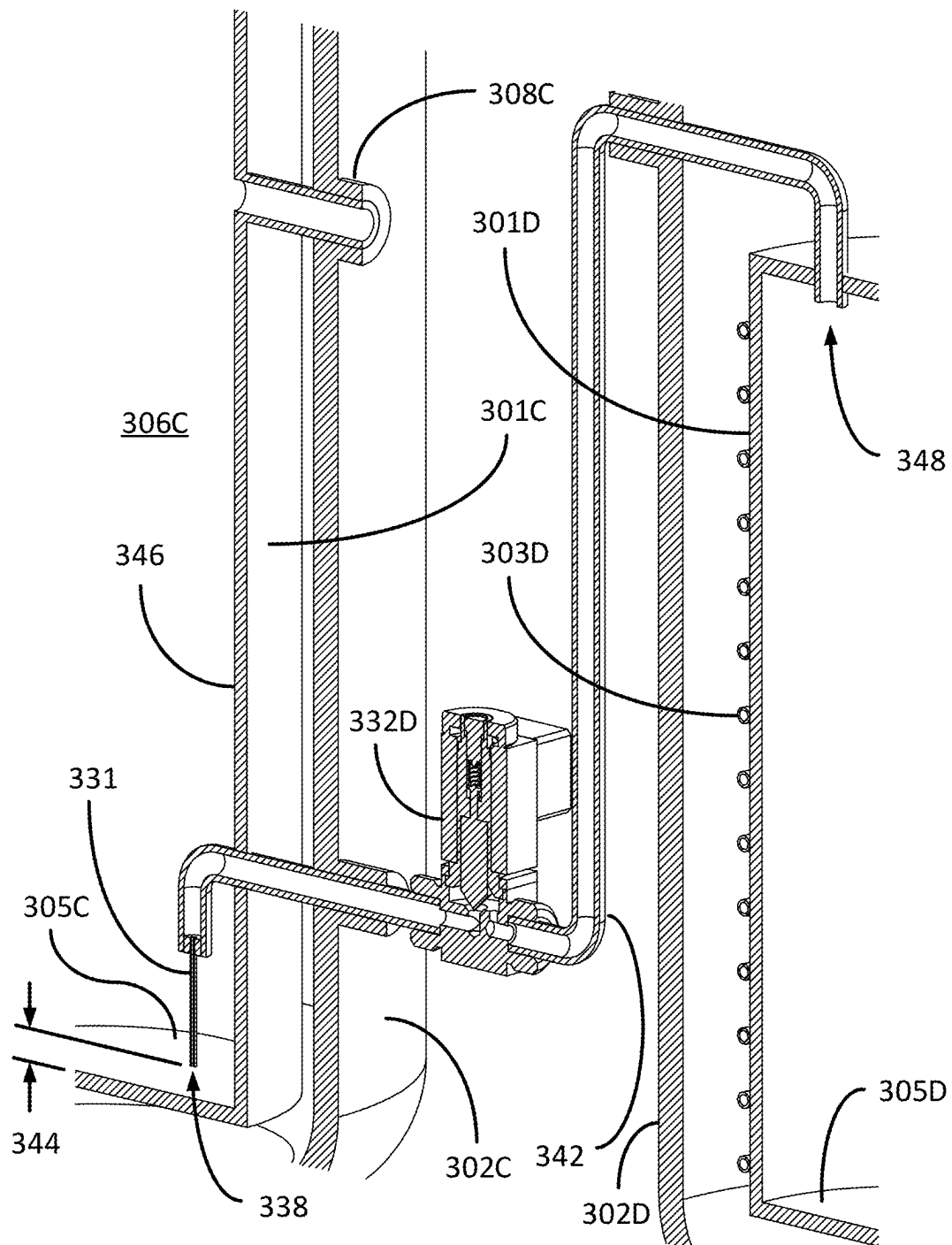


FIG. 6E

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PROGRESSIVELY CHILLED GAS SEPARATION TANKS

CROSS-REFERENCE TO RELATED APPLICATIONS

This is a continuation-in-part application which claims priority to and the benefit of U.S. patent application Ser. No. 18/894,605 entitled: IONIZED GAS SEPARATION ARRANGEMENT filed on Sep. 24, 2024, the entire disclosure of which is hereby incorporated by reference, which is a continuation-in-part application claiming priority to and the benefit of U.S. patent application Ser. No. 18/766,818 entitled: CRYOGEN CHAMBER WITH ADSORBER, filed on Jul. 9, 2024.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates generally to mining extraterrestrial sites for gas trapped in extraterrestrial soil.

2. Description of Related Art

Helium-3 (He-3) holds immense potential as a clean and efficient energy source, offering numerous benefits for various applications. It has been discovered that He-3 is a highly efficient fuel for nuclear fusion, which has the potential to provide a nearly limitless and environmentally friendly energy solution. When He-3 fuses with deuterium, it produces a helium atom, a proton, and a tremendous amount of energy. Unlike traditional nuclear fission, fusion reactions release energy without generating hazardous radioactive waste or greenhouse gas emissions. Harnessing He-3 for fusion could revolutionize the energy landscape, providing a stable and sustainable source of power for the future.

However, despite its remarkable benefits, mining He-3 on Earth faces significant limitations. For one thing, He-3 is incredibly scarce on our planet. It is primarily found in minute quantities in the top surfaces of lunar soil, known as regolith, and is sparsely present in the Earth's atmosphere. Consequently, the extraction of He-3, whether on Earth or the Moon, is challenging and expensive. Moreover, mining it on the Moon poses logistical difficulties requiring advanced technologies and significant investments. Even assuming mining on the Moon is successful, transporting the mined He-3 from the Moon to Earth presents significant technical challenges and high costs. These limitations highlight the need for further research and development to find more accessible and cost-effective sources of He-3 or to explore alternative fusion fuel options that do not rely solely on He-3.

The subject matter disclosed herein is generally directed to innovations related to collecting He-3 and other useable gaseous elements on extra-terrestrial bodies.

SUMMARY OF THE INVENTION

The present invention generally relates to mining extraterrestrial sites for gas trapped in extraterrestrial soil.

In that light, certain embodiments of the present invention envision a gas separation arrangement comprising a chamber connected to a plurality of progressively chilled separation tanks that successively separate out different species of gas. The chamber comprises at least He-3 gas and He-4 gas. A first separation tank is linked to the chamber via a first

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transfer tube and a second separation tank is in liquid communication with the first separation tank A via a capillary transfer tube. Liquid communication means that liquid is in contact with the elements the liquid flows through or wherein the elements directly facilitate the movement of liquid. For example, though liquid may flow through a transfer tube that passes through an outer tank housing, it should be appreciated that the outer tank housing is not in liquid communication with the fluid by virtue of the transfer tube simply passing through it. A first heat exchanger comprises a first He-4 supply, the first He-4 supply is configured to maintain the first separation tank at a first tank temperature below a He-4 liquification temperature. A second heat exchanger comprises a second He-4 supply, the second He-4 supply is configured to maintain the second separation tank at a second tank temperature below the He-4 liquification temperature. A first collection port is in fluid communication with a first interior of the first separation tank, wherein the first collection port is configured to collect the He-3 gas. A second collection port is in fluid communication with a second interior of the second separation tank, wherein the He-4 gas configured to be collected from the second collection port.

Yet other embodiments of the present invention contemplate a gas separation apparatus comprising a chamber connected to a plurality of progressively chilled separation tanks that successively substantially separate out different species of gas. The apparatus can therefore comprise a first separation tank linked to a chamber via a first transfer tube and a second separation tank in liquid communication with the first separation tank A via a capillary transfer tube having a capillary inlet port within 0.5 inches of a bottom surface of the first separation tank. A first heat exchanger is coupled to the first separation tank, wherein the first heat exchanger is configured to cool the first separation tank below a first temperature where He-4 liquifies. A second heat exchanger is coupled to the second separation tank, wherein the second heat exchanger is configured to cool the second separation tank below the first temperature. A first collection port is in fluid communication with a first interior of the first separation tank, the first collection port is configured to collect He-3 gas. A second collection port is in fluid communication with a second interior of the second separation tank, the second collection port is configured to collect He-4 gas.

Still other embodiments of the present invention contemplate a method to substantially separate and obtain He-3 and the He-4. The method comprising a step for dispensing the He-3 and the He-4 in a gas form into a first separation tank from a gas collection chamber; a step for cycling a first He-4 supply at a first temperature through a first heat exchanger that is connected to the first separation tank; a step for cooling the first separation tank to a first temperature that is below a liquification temperature of the He-4 via the first He-4 supply; a step for liquifying the He-4 in the first separation tank; a step for transporting the liquified He-4 into a second separation tank via a capillary tube that connects the first separation tank to the second separation tank; and a step for collecting the liquified He-4 from the second separation tank via a second collection port.

Certain other embodiments of the present invention envision a gas separation arrangement that is generally directed to an ion diverter that is used to segregate and capture the segregated gas from an adsorption gas capturing chamber (chamber). The arrangement can comprise a chamber containing at least a first species of gas. The chamber is connected to an ion diverter via a gas transmission port. The ion diverter generally comprises a valve configured to

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control gas communication between the ion diverter and the chamber. The ion diverter includes an ionizer adjacent to the gas transmission port, an ion deflector comprising a first magnet plate separated from a second magnet plate by a separation spacing defining a channel, and an ion accelerator between the ionizer and the ion deflector. A plurality of target receptacles is disposed at an exit location of the channel, wherein each target receptacle corresponds to a different target gas. The ionizer is configured to ionize the first species of gas and the second species of gas received from the gas transmission port. The ion deflector is configured to direct the ionized first species of gas in a first controlled trajectory from the ion accelerator to the first target receptacle and the ionized first species of gas in a second controlled trajectory from the ion accelerator to the second target receptacle.

Another embodiment of the present invention envisions a gas separator comprising a gas transmission port that is interposed between an ionizer and a chamber, wherein the chamber is configured to contain a first species of gas and a second species of gas. Connected to the chamber and ionizer is an ion deflector that comprises a pair of magnetic plates, which are separated by a channel. There is an accelerator between the ionizer and the ion deflector and a valve configured to control gas communication between the chamber and the ionizer. There is a first and a second target receptacle and disposed at a first and a second exit location, respectively, of the channel. The ionizer is configured to ionize the first species of gas that is received from the gas transmission port and further configured to ionize the second species of gas received from the gas transmission port. The accelerator is configured to accelerate the ionized first species of gas and the ionized second species of gas towards the ion deflector. The ion deflector is configured to deflect the ionized first species of gas in a first curved trajectory between the accelerator and the first target receptacle and is configured to deflect the ionized second species of gas in a second curved trajectory between the accelerator and the second target receptacle.

Yet another embodiment of the present invention envisions a gas separator device comprising a gas transmission pathway having a second part and a first part that extends from a chamber through a gas transmission port before reaching an ionizer and ion accelerator, after which (the ion accelerator) the second part is arched through a magnetic field before terminating at a target receptacle. The chamber is configured to contain a first gas species. The ionizer is configured to ionize the first gas species. The accelerator is configured to accelerate the ionized first gas species towards the magnetic field. The magnetic field is established between a pair of separated magnetic plates, wherein the magnetic field is configured to bias the ionized first gas species along the second part to the target receptacle, which is configured to trap the ionized first gas species.

Still another embodiment of the present invention contemplates a method comprising providing a chamber that is connected to an ion diverter. The method further envisions flowing a first gas species and a second gas species from the chamber to the ion diverter via a gas transmission port, followed by ionizing the first gas species and the second gas species in an ionizer, followed by accelerating the ionized first gas species and the ionized second gas species, via an ion accelerator, into a magnetic field generated by a pair of spaced apart magnetic plates and. The ionized first gas species is biased in a first arc, via the magnetic field, to a first target receptacle and the ionized second gas species is biased in a second arc, via the magnetic field, to a second target

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receptacle. The ionized first gas species is directed to and captured in the first target receptacle and the ionized second gas species is directed to and captured in the second target receptacle.

Other embodiments of the present invention envision a gas collection system generally comprising a gas segregation chamber, at least one cooling plate in the gas segregation chamber, an adsorption gas capturing chamber connected to the gas segregation chamber and a carbon adsorber in the adsorption gas capturing chamber. The gas segregation chamber comprises a housing, wherein the housing is defined by housing sides that extend from a top housing surface to a rim. A first interior environment is defined within the housing, wherein the first interior environment is in communication with an external environment through only the rim. The external environment being defined outside of the housing. At least one cooling plate is in the gas segregation chamber, wherein the least one cooling plate comprises a passageway that is configured to accommodate cryogenic fluid. An adsorption gas capturing chamber is connected to the housing, wherein the adsorption gas capturing chamber comprises a second interior environment that is in communication with the first interior environment via a connecting port. A carbon adsorber is in the second interior environment.

In another aspect of the present invention, some embodiments envision a gas collection arrangement that generally comprises a gas segregation chamber, at least one cooling plate in the gas segregation chamber, an adsorption gas capturing chamber and a carbon adsorber. The gas segregation chamber defines a first interior environment when a rim of the gas segregation chamber rests atop regolith. The gas segregation chamber also comprises at least one cooling plate that is configured to capture higher temperature condensing gas but not low temperature condensing gas, wherein the gases are released from the regolith. The adsorption gas capturing chamber defines a second interior environment that is in communication with the first interior environment via a connecting port. The carbon adsorber is in the second interior environment, wherein the carbon adsorber is configured to capture the low temperature condensing gas.

Another embodiment of the present invention contemplates a segregating gas arrangement generally comprising a gas segregation chamber, at least one cooling plate in the gas segregation chamber, at least one cooling plate in the gas segregation chamber, and a carbon adsorber. The gas segregation chamber comprises a rim that when resting atop regolith defines a first interior environment. The at least one cooling plate is in the gas segregation chamber, wherein the least one cooling plate is maintained at a first temperature above 5° K, which is a condensation temperature at which higher temperature condensing gases condense. The adsorption gas capturing chamber defines a second interior environment that is in communication with the first interior environment. The carbon adsorber is in the second interior environment and is maintained at a second temperature below 3° K. The carbon adsorber is configured to capture the low temperature condensing gas.

Certain other embodiments of the present invention envision a gas segregating method comprising providing a segregating gas arrangement comprising an adsorption gas capturing chamber that is connected to a gas segregation chamber. The gas segregation chamber comprises a housing that is defined by housing sides that extend from a top housing surface to a rim. The method further comprises resting the rim atop regolith, wherein a first interior envi-

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ronment is defined within the housing when the rim is resting atop the regolith. A first temperature above 5° K is maintained in at least one cooling surface that is disposed in the gas segregation chamber. A majority of higher temperature condensing gases are condensed in the first interior environment but not a lower temperature condensing gas is not condensed in the first interior environment. The lower temperature condensing gas is captured in a carbon adsorber that is located in the adsorption gas capturing chamber, wherein the lower temperature condensing gas migrates from the first interior environment to a second interior environment that is defined within the adsorption gas capturing chamber.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are line drawings of isometric views of the gas capturing arrangement depicting the base open and closed consistent with embodiments of the present invention;

FIG. 1C is a line drawing of a side view of the gas capturing arrangement consistent with embodiments of the present invention;

FIG. 1D is a pseudo cross-section line drawing of the gas capturing arrangement consistent with embodiments of the present invention;

FIG. 1E is a line drawing of a side view cross-section of the gas capturing arrangement from the side view of FIGS. 1C and 1D consistent with embodiments of the present invention;

FIG. 2A is a line drawing of a cross-section view of the gas segregation chamber consistent with embodiments of the present invention;

FIG. 2B is a line drawing of a higher resolution cross-section line drawing of the cooling plates consistent with embodiments of the present invention;

FIG. 3 is a higher resolution cross-section line drawing of the adsorption gas capturing chamber and cryogenic liquid coolant reservoir consistent with embodiments of the present invention;

FIG. 4 is a block diagram describing method steps of using the segregating gas arrangement consistent with embodiments of the present invention;

FIG. 5A is a line drawing of the gas separation arrangement connected to the adsorption gas capturing chamber consistent with embodiments of the present invention;

FIG. 5B is a line drawing of a cross-section of the gas separation arrangement depicting the gas transmission pathway in the gas separation arrangement;

FIG. 5C is a line drawing depicting the gas separation arrangement in a cross-section side view consistent with embodiments of the present invention;

FIG. 5D is a line drawing cross-section of the segment between the chamber and ion diverter consistent with embodiments of the present invention;

FIG. 5E is a line drawing depicting a cross-section of the receptacle arrangement consistent with embodiments of the present invention;

FIG. 6A is a line drawing depicting a cross-section of a front view of a gas separation and collection arrangement consistent with embodiments of the present invention;

FIG. 6B is a line drawing depicting a top view of the gas separation arrangement consistent with embodiments of the present invention;

FIG. 6C is a line drawing depicting an angled semi-cross-section view of a gas separation unit consistent with embodiments of the present invention;

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FIG. 6D is a line drawing depicting an isometric view of another gas separation arrangement consistent with embodiments of the present invention; and

FIG. 6E is a line drawing depicting an isometric view of a portion of the third separation tank and the fourth separation tank cross sectioned consistent with embodiments of the present invention.

DETAILED DESCRIPTION

Initially, this disclosure is by way of example only, not by limitation. Thus, although the instrumentalities described herein are for the convenience of explanation, shown and described with respect to exemplary embodiments, it will be appreciated that the principles herein may be applied equally in other similar configurations involving the subject matter directed to the field of the invention. The phrases “in one embodiment”, “according to one embodiment”, and the like, generally mean the particular feature, structure, or characteristic following the phrase, is included in at least one embodiment of the present invention and may be included in more than one embodiment of the present invention. Importantly, such phrases do not necessarily refer to the same embodiment. If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic. As used herein, the terms “having”, “have”, “including” and “include” are considered open language and are synonymous with the term “comprising”. Furthermore, as used herein, the term “essentially” is meant to stress that a characteristic of something is to be interpreted within acceptable tolerance margins known to those skilled in the art in keeping with typical normal world tolerance, which is analogous with “more or less.” For example, essentially flat, essentially straight, essentially on time, etc. all indicate that these characteristics are not capable of being perfect within the sense of their limits. Accordingly, if there is no specific +/- value assigned to “essentially”, then assume essentially means to be within +/-2.5% of exact. The term “connected to” as used herein is to be interpreted as a first element physically linked or attached to a second element and not as a “means for attaching” as in a “means plus function”. In fact, unless a term expressly uses “means for” followed by the gerund form of a verb, that term shall not be interpreted under 35 U.S.C. § 112(f). In what follows, similar or identical structures may be identified using identical callouts.

With respect to the drawings, it is noted that the figures are not necessarily drawn to scale and are diagrammatic in nature to illustrate features of interest. Descriptive terminology such as, for example, upper/lower, top/bottom, horizontal/vertical, left/right and the like, may be adopted with respect to the various views or conventions provided in the figures as generally understood by an onlooker for purposes of enhancing the reader's understanding and is in no way intended to be limiting. All embodiments described herein are submitted to be operational irrespective of any overall physical orientation unless specifically described otherwise, such as elements that rely on gravity to operate, for example.

Described herein are embodiments directed to collecting vaporize gaseous atoms and molecules in an extremely low-pressure environment and collecting those vaporized gaseous atoms and molecules using cryogenically cooled surfaces, such as plates. Extremely low-pressure environment is defined herein as below 1×10^{-5} bars, wherein pressure at sea-level on Earth is approximately 1 bar.

Aspects of the present invention consider mining gaseous atoms and molecules on extraterrestrial bodies such as the Moon, asteroids, moons orbiting other planets, etc., for example. Many of these extraterrestrial bodies have little to no ambient pressure at or just beyond their surfaces and depending on the size of the extraterrestrial body have a significantly lower gravitational pull than that experienced on Earth.

Though embodiments of the present invention can be used in conjunction with many different extraterrestrial bodies, it is one object of the present invention to focus on mining gaseous atoms and molecules (materials) from the Moon. The Moon's atmosphere (the surface boundary exosphere) has a pressure of about 3×10^{-15} bars and can range in temperature between 20° to 400° Kelvin. In the interest of continued exploration of the Moon as well as maintaining long-term residents on the Moon, extracting or otherwise mining important gaseous materials from the Moon, such as oxygen, nitrogen, hydrogen, and helium, for example, reduces the dependency of transporting such gaseous materials from the Earth. Moreover, helium-3, a light stable isotope of helium having two protons and one neutron, which has promise as an important constituent in fusion reactions, is far more abundant on the Moon than the Earth. In some estimates, helium-3 is more than a thousand times more abundant on the Moon than compared to the Earth making the Moon a better target to obtain helium-3.

Certain embodiments of the present invention envision extracting target gaseous materials from the Moon by heating up moon regolith (lunar soil/minerals) to gas vaporizing temperatures defined as temperatures that are high enough to liberate/vaporize these target gaseous materials from moon regolith, or simply "regolith". The vaporized target gaseous materials are then collected as liquid from condensation surfaces that are at or below the condensation temperatures corresponding to each of the target gaseous materials. When condensed i.e., liquefied, the liquid or frozen, which improves the transportation of these target materials.

In that light, some embodiments of the present invention contemplate a vapor collection system that segregates higher temperature condensing vapor, such as hydrogen, oxygen, and nitrogen, from lower temperature condensing vapor, such as helium, that can be used at an extra-terrestrial body to collect target gaseous atoms and molecules that are floating around in a shielded environment at a pressure at or less than 1×10^{-5} bar.

Presented below are also embodiments of a segregating gas arrangement that generally comprises a gas segregation chamber, at least one cooling plate in the gas segregation chamber, and a carbon adsorber in an adsorption gas capturing chamber. The gas segregation chamber has a rim that when resting atop regolith defines a first interior environment. The cooling plates are in the gas segregation chamber, wherein the cooling plates are maintained at a first temperature above 5° K, which is a condensation temperature that higher temperature condensing gases will condense. The adsorption gas capturing chamber defines a second interior environment that is in communication with the first interior environment. The carbon adsorber is in the second interior environment and is maintained at a second temperature below 3° K. The carbon adsorber is configured to capture the low temperature condensing gas.

Also presented below are embodiments that disclose a gas separation arrangement for separating and containing various types of gas received from a collection chamber. In certain embodiments, the gas separation arrangement is envisioned to assist in mining select gases from the Moon.

The arrangement includes plurality of progressively chilled separation tanks that successively substantially separate out different species of gas (such as 80% separation). The separation tanks are connected to and receive the various gas species from the collection chamber. Each separation tank can have an independent heat exchanger that maintains the respective separation tank at a temperature that selectively liquifies/precipitates out a target species of gas allowing the remaining gas to move to a successive separation tank. Ultimately, He-3 and He-4 are separated from one another to be collected, like the other gas species.

With respect to FIGS. 1A and 1B, the reservoir **160** is shown connected to and supported by the gas capturing arrangement base **104** via support legs **101**, however a skilled artisan will immediately appreciate that there are numerous other ways of supporting a reservoir **160** to the gas capturing arrangement **100** without departing from the scope and spirit of the present invention. The carbon adsorber **145** and in some cases the entire adsorption gas capturing chamber **140** can be removed by an adsorber access arm **180** via an access port **165** in the reservoir **160**, which would be done after the upper gate valve **124** closes off the connecting passageway **125**. A sliding gas segregation chamber gate valve gate **190** (door) is interposed between the gas capturing arrangement base **104** and a rim **120** that is arranged and configured to rest atop a granular surface **200**, such as regolith. The sliding gas segregation chamber door **190** comprises an inlet aperture **192** that is aligned with an intake port **121** defined as the space within the inside boundary of the rim **120**. The sliding gas segregation chamber door **190** can be actuated by a motor **196** that can be electrically connected to a power source (not shown) via an electrical connector **198**. It should be appreciated that though motors, electronics, computers, algorithms may not be shown, such elements can be employed to enhance the functionality of the embodiments described below, however their absence does not change the fundamental functionality of the embodiments to enable the reader to appreciate the scope of the ideas presented herein.

As shown in FIG. 1C in view of FIGS. 1D and 1E, the gas capturing arrangement **100**, and more particularly, the rim **120** is resting on or otherwise in contact with regolith **200**. The gas segregation chamber gate valve **190** is open providing communication between the internal segregation chamber environment **112** and the regolith **200** via the inlet region **121**, as shown in FIG. 1B. The adsorption gas capturing chamber **140** is connected to the top housing surface **118** of the gas segregation chamber **110** via the connecting passageway **125**. The gate **124** of the upper gate valve **126** is open thereby providing communication between the internal segregation chamber environment **112** and the adsorption gas capturing chamber environment **142**. When the upper gate valve **126** is closed, the adsorption gas capturing chamber **140** can be separated from the gas segregation chamber **110** at the separation junction **128**. The adsorption gas capturing chamber **140** can be lifted from the gas segregation chamber **110** through the access port **165** in the reservoir **160** via the adsorber access arm **180**. Liquid cryogen is delivered from the reservoir **160** to the gas segregation chamber **110** via lines in the feed and return line conduit **175** and to the upper chamber heat sink **144** via the lines in upper chamber conduit **146**. In the present embodiment, the cryogenic liquid coolant reservoir **160** is held in place relative to the base **104** via the support legs **101**.

FIG. 1D is a pseudo cross-section line drawing of the gas capturing arrangement **100** consistent with embodiments of the present invention. Notably, the adsorber **145** is shown in

the upper chamber 140. The adsorber 145 is wrapped around and in contact with a heat sink 144, such as a copper block, that is cryogenically cooled by liquid cryogen circulated through feed and return lines 148, the liquid cryogen contained in the cryogenic liquid coolant reservoir 160. The gate valve 126 is configured to close off or open gas communication between the upper and lower chambers 140 and 110. Gas communication is defined as the ability of gas to move from one element or location into another. Cryogenic feed line 172 provides liquid cryogen to the cooling plates 105 in the gas segregation chamber 110 where gas liberated from regolith 200 (via a heater 150 heating the regolith 200) entering the internal segregation chamber environment 112 via the inlet region 121. The liquid cryogen is provided by the cryogenic liquid coolant reservoir/s 160 and circulated through the cooling plates 105 via the feed and return lines 172 and 174.

FIG. 1E is a block diagram of the gas capturing arrangement 100 of FIG. 1D depicting the cross-section of the gas capturing arrangement 100 consistent with embodiments of the present invention. The gas segregation chamber 110 can be configured and arranged like a cryopump with one or more plates 105 that separate out certain types of gases released from the regolith 200 from entering the upper chamber 140. In some embodiments, select gas types can enter the upper chamber, such as H-1, H-2, He-3, He-4, for example.

With more specificity, one embodiment contemplates a plurality of plates 105 with an upper first plate or plates 106 being at a first temperature, a middle second plate or plates 107 maintained at a second temperature, and a lower third plate or plates 108 maintained at a third temperature. This embodiment contemplates the option of the first temperature being colder than the second temperature, the second temperature being colder than the third temperature. In other words, the plates 105 get progressively colder as they near the top housing surface 118. The first temperature can be maintained by pumping a first cryogenic fluid through lines in the first plate or plates 106. The second temperature can be maintained by pumping a second cryogenic fluid through lines in the second plate or plates 107. The third temperature can be maintained by pumping a third cryogenic fluid through lines in the third plate or plates 108. Certain embodiments contemplate the first cryogenic fluid being liquid helium, the second cryogenic fluid being liquid hydrogen, and the third cryogenic fluid being liquid nitrogen.

Another embodiment contemplates one or more plates 105 in the gas segregation chamber 110 being infused with a third cryogenic fluid at a third temperature to segregate out a first gas, such as water vapor, that will condense and freeze on the surface of the one or more plates 105. After a first period of time that is sufficient to segregate out the first gas, the one or more plates 105 can be infused with a second cryogenic fluid at a second temperature to segregate out a second gas, such as nitrogen, that will condense on the surface of the one or more plates 105. After a second period of time that is sufficient to segregate out the second gas, the one or more plates 105 can be infused with a first cryogenic fluid at a first temperature to segregate out a third gas, such as oxygen and hydrogen, that will condense on the surface of the one or more plates 105. The first cryogenic fluid is colder than the second cryogenic fluid, which is colder than the third cryogenic fluid. Certain embodiments contemplate the first cryogenic fluid being liquid helium, the second cryogenic fluid being liquid hydrogen, and the third cryogenic fluid being liquid nitrogen.

Yet another embodiment contemplates the one or more plates 105 in the gas segregation chamber 110 being infused with a single cryogenic fluid, such as liquid helium, that is configured to segregate all gasses out of the gas segregation chamber 110 except helium. In this embodiment, the one or more plates 105 can be maintained at a temperature slightly above the condensation temperature of helium, such as via heat boosts or design of the cryogenic fluid carrying passageways 122 in the cooling plates 105.

The gas segregation chamber 110 is defined within a segregation chamber housing 115. The segregation chamber housing 115 is defined by housing sides 116 that extend from a top housing surface 118 to a rim 120. The segregation chamber housing 115 defines an internal environment 112, which is in communication with an external environment 102 via an inlet region 121 at the rim 120 when the lower gate valve 190 is open. The inlet region 121 is an opening that spans that the rim 120. For example, if the rim 120 is circular with a radius r , the inlet region 121 is the area of the circle (πr^2). This assumes the inlet aperture 192 in the lower gate valve 190 is open or otherwise not obstructing the inlet region 121. The external environment 102 is defined outside of the housing 115.

The rim 120 is configured and arranged to rest atop a granular surface 200, such as regolith. There can be a filter 130 that is configured to trap regolith dust from entering the internal environment 210 via the inlet region 121. The filter 130 can be a HEPA filter or ULPA filter, for example. Certain embodiments envision to filter but rather the use of magnets with or without ionizers to electromagnetically prevent the regolith dust from entering the internal segregation chamber environment 112.

The gas capturing arrangement 100 can further comprise a heating element 150 configured to emit energy 152 that heats a target volume 154 of regolith 200 at and beyond the rim 120. The heating element 150 can be a laser that emits a laser beam, an ultrasound generator that emits ultrasound, a microwave generator that emits microwaves, a radiant heater that emits radiant heat, etc. When the regolith 200 is heated, gaseous elements are liberated and captured within the gas segregation chamber 110. Some embodiments envision the heating element not being in the gas segregation chamber 110 but rather outside of the gas segregation chamber 110, wherein the externally located heating element directs heat in the regolith 200 under the gas segregation chamber 110.

The adsorption gas capturing chamber 140 is attached to the gas segregation chamber 110 via a connecting passageway 125, which provides communication between the two chambers 110 and 140. In this embodiment, communication can be broken between the internal segregation chamber environment 112 the adsorption gas capturing chamber environment 142 via the gate valve 126 when the gate 124 is closed. Certain embodiments contemplate the adsorption gas capturing chamber 140 being detachable from the gas segregation chamber 110, such as for example at the gate valve 126. Other embodiments contemplate the adsorption gas capturing chamber 140 being directly attached to the gas segregation chamber 110 (that is with no connecting passageway 125 therebetween).

With respect to the adsorption gas capturing chamber 140, disposed therein is a carbon adsorber 145 configured and arranged to capture isolated gas that is not targeted to be captured in the gas segregation chamber 110. The carbon adsorber 145 is comprised of a highly porous carbon that is well known for having a high surface area due to its high porosity. Certain embodiments envision capturing the iso-

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lated gas that migrates into the adsorption gas capturing chamber **140** in the pores of the carbon adsorber **145**. In the present embodiment, the carbon adsorber **145** is in contact with a heat sink **142**, which is maintained at or near the cryogenic temperature of the cryogenic fluid circulating therethrough via the upper chamber cryogen feed and return lines **148**. The heat sink **142** is a body at essentially the lowest temperature in the gas capturing arrangement **100**. The upper chamber heat sink **144** cools the carbon adsorber **145** by way of conduction to near the temperature of the heat sink **142**, within a degree or two Kelvin.

The adsorption gas capturing chamber **140** is connected to an adsorber access arm **180** that can assist in removing the adsorption gas capturing chamber **140** from the gas capturing arrangement **100**. Certain embodiments envision a duct **182** inside of the adsorber access arm **180** configured to pull gas collected in the carbon adsorber **145** for storage and transport via a secondary gas receiving system (not shown).

FIG. 2A is a line drawing of a cross-section view of the gas segregation chamber **110** consistent with embodiments of the present invention. FIG. 2A is in view of FIG. 1C identifying associated elements called out. In this embodiment, there are six cooling plates **105** attached to one another via a stem **103** extending through the center of the cooling plates **105**. The stem **103** can attach to the segregation chamber housing **115** via a web or spokes, not shown, which permit unobstructed movement of gas from the gas segregation chamber **110** into the adsorption gas capturing chamber **140** via the connecting passageway **125**. Each of the cooling plates **105** comprise cryogenic fluid carrying passageways **122** through which cryogenic fluid is circulated from the feed line **172**. The cryogenic fluid carrying passageways **122** ultimately loop back to the cryogenic fluid coolant reservoirs **160** via the return line **174**. The feed line **172** and return line **174** are shielded by a feed and return line conduit **175**. The cryogenic fluid is circulated through the cryogenic fluid carrying passageways **122** in the cooling plates **105** to bring the temperature of the cooling plates to below the condensation temperature of one or more target gasses in the internal segregation chamber environment **112**. In the present embodiment, the cooling plates **105** are disk shaped but other embodiments contemplate other shapes, such as blades, rectangles, etc., without departing from the scope and spirit of the present invention. The gas segregation chamber **110** is considered a cryogenic 'roughing pump' to segregate out a majority of higher temperature condensing gasses from the lower temperature condensing gases before the lower temperature gasses migrate into the adsorption gas capturing chamber **140**.

In practice, the heating element **150** heats the regolith **200** to a temperature above which will liberate gasses trapped in the regolith **200**. The liberated gasses (of which there may be several different types, such as hydrogen, oxygen, helium, nitrogen, argon, etc.) enter the internal segregation chamber environment **112** via the inlet region **121** (when open) defined as either the area within the rim **120** or the area of the inlet aperture **192**, whichever is smaller. In some embodiments, the inlet area within the rim **120** is the same as the area of the inlet aperture **192**. The heating element **150** can diffuse energy to the surface of the regolith **200**. The energy can be radiant energy, microwave energy, ultrasonic energy, conductive energy from a contact heater, or a laser that sweeps over an area under the area within the rim **120**, just to name several non-limiting examples. In this embodiment a filter **130** interposed between the rim **120** and the cooling plates **105** prevents or at the least greatly reduces

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any dust from entering the internal segregation chamber environment **112** without blocking any gasses.

With continued reference to the cooling plates **105**, certain embodiments envision the cooling plates **105** being maintained at a temperature that is cold enough to condense out all gasses except for helium. In this case, liquid helium is circulated through the cooling plates **105** to maintain a temperature of the cooling plates **105** slightly above the condensation temperature of helium. In this way, helium is segregated from the other gases and will migrate through the connecting passageway **125** and into the adsorption gas capturing chamber **140**, assuming the upper gate valve **126** is open. Meanwhile, the cooling plates **105** will, condensed water, oxygen, nitrogen, hydrogen, etc. hence the roughing pump portion of the gas collection arrangement **100**. The valve **126** can be closed and the adsorption gas capturing chamber **140** can be separated at separation junction **128** for processing the contents of each chamber **110** and **140** remotely or on-site.

Certain other embodiments envision multiple different kinds of cryogenic fluid at different temperatures either made to flow through the cooling plates **105** successively to provide a successively colder internal segregation chamber environment **112** as discussed above. For example, liquid nitrogen can be made to flow through the cooling plates **105** first, followed by liquid oxygen, then liquid hydrogen and lastly followed by liquid helium. Another embodiment envisions the bottom two cooling plates **105** being maintained at a temperature that condenses water, the middle two cooling plates **105** being held at a temperature to condense oxygen and the upper two cooling plates **105** being held at a temperature that condenses hydrogen. This can be accomplished with multiple fluid cryogenics or a single cryogen with different geometries of the cryogenic fluid carrying passageways **122** in each set of cooling plates **105** or optionally electric heaters embedded in the cooling plates **105** for temperature control, just to name several examples. The condensate on the cooling plates **105** can be collected and processed for later use. The heaters in the cooling plates **105** can further be used to heat the cooling plates **105** to a temperature the liberates the condensed gas from the plate surfaces for further processing.

FIG. 2B is a higher resolution cross-section line drawing of the cooling plates **105** consistent with embodiments of the present invention. As shown, there is a downward arrow indicating the direction of cryogenic fluid flow in the feed line **172**. The feed line **172** connects into each cooling plate **105** via the cryogenic fluid carrying passageways **122**. In the present embodiment, the cryogenic fluid carrying passageways **122** in each cooling plate **105** connect to one another through the stem **103**. The cryogenic fluid carrying passageway **122** in the bottom cooling plate **105** connects to the return line **174** in the stem **103** where it can flow (be pumped) back into the reservoir tank **160**, shown by the up facing arrow.

FIG. 3 is a higher resolution cross-section line drawing of the adsorption gas capturing chamber **140** and cryogenic liquid coolant reservoir **160** consistent with embodiments of the present invention. As shown, a pump **158** disposed in the reservoir **160** is configured to circulate cryogenic liquid coolant through the upper chamber cryogen feed and return lines **148** and in one embodiment the lower chamber cryogen feed line **172** and return line **174**. Other embodiments envision a separate pump circulating cryogenic liquid coolant through the lower chamber cryogen feed line **172** and return line **174**. The adsorption gas capturing chamber **140** comprises a carbon adsorber **145** that is in contact with the

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upper chamber heat sink **144**. The carbon adsorber **145** is cooled through conduction while the heat sink **144** is being cooled by the cryogen circulating through the upper chamber cryogen feed and return lines **148**. In operation, the adsorption gas capturing chamber **140** is envisioned to mostly contain helium gas (as the segregated, targeted gas) since the other gasses liberated from the regolith **200** are mostly condensed in the gas segregation chamber **110**. The helium gas contains a concentration of both He-3 and He-4 of which become trapped in the carbon adsorber **145**. Because He-3 condenses at around 3° K, the carbon adsorber **145** is kept below that temperature to help retain He-3 and He-4.

The contents of the carbon adsorber **145**, such as He-3 and He-4 from the example above, can be harvested by closing the gate valve **126**, separating the adsorption gas capturing chamber **140** from the connecting passageway **125** at the separation junction **128**, and removing the adsorption gas capturing chamber **140** through the access port **165** in the reservoir **160**. Once removed, the adsorption gas capturing chamber **140** can be replaced with a new or ready to use adsorption gas capturing chamber **140**. The adsorption gas capturing chamber **140** that has been collecting gas can be processed in a processing facility equipped to harvest or otherwise extract the gas captured in the adsorption gas capturing chamber **140** and the carbon adsorber **145**. Optionally, the gas captured in the adsorption gas capturing chamber **140** and the carbon adsorber **145** can be extracted without removing the adsorption gas capturing chamber **140** via the access arm duct **182** extending through the adsorber access arm **180** by connecting an extractor at the access channel **184** at the top of the adsorber access arm **180**. In yet another harvesting option, the carbon adsorber **145** is envisioned to be a cartridge that is removable from the adsorption gas capturing chamber **140** and replaceable with a new or ready to use cartridge.

FIG. 4 is a block diagram describing method steps of using the segregating gas arrangement **100** consistent with embodiments of the present invention. FIG. 4 is intended to be seen in view of the preceding figures with the callout numbers corresponding to the elements referred to therein. Step **302** refers to the segregating gas arrangement **100** as described above. Though the adsorption gas capturing chamber **140** is shown connected to the top portion of the gas segregation chamber **110**, the adsorption gas capturing chamber **140** can be connected elsewhere. Certain other embodiments envision a single gas segregation chamber **110** (no separate adsorption gas capturing chamber **140**) with the carbon adsorber **145** therein using potentially separate temperatures at different times to capture higher condensing temperature gasses then lower condensing temperature gasses as described herein.

Step **204** is a step for locating the segregating gas arrangement **100** to a spot/location on the Moon or some other extraterrestrial body and resting the rim **120** of the gas segregation chamber **110** atop regolith **200**. The positioning could be accomplished with a rover, manually, by a hovering device, or simply by deploying the segregating gas arrangement **100** from an orbiting or hovering craft.

Once deployed with the rim **120** resting on the regolith **200**, as shown in step **306**, the cooling plates **105** in the gas segregation chamber **110** are brought down to a temperature at or above 5° K, which can be accomplished by circulating liquid helium through channels **122** in the cooling plates **105**. The cooling plates **105** are one embodiment of a cooling surface, which could be one or more screens, blades, channels, or some other surface that can be chilled as

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understood by those skilled in the art. 5° K is the temperature that all higher condensing temperature gasses other than helium will condense, which will essentially segregate or otherwise rough the higher temperature condensing gasses out in the gas segregation chamber **110**. This process isolates helium gas (He-3 and He-4) to migrate into the adsorption gas capturing chamber **140** where it can be adsorbed by a carbon adsorber **145** maintained at below 3° K, which is a temperature that helium condenses. In order to accomplish these low temperatures, the liquid helium supplied by the reservoir **160** can be pressurized and further cooled. This process can be adjusted to different temperatures to target specific gasses having a higher condensing temperature than helium, such as targeting hydrogen or something else while roughing out even higher condensing temperature gasses.

As presented in step **308**, the gas to be captured within the segregating gas arrangement **100** is liberated from the regolith **200** under the rim **120** via a heater **150** that heats up the regolith **200** to temperatures that may exceed 800° K. The liberated gas enters the interior environment **112** of the gas segregation chamber **110** through an opening (inlet region) **121** in the rim **120**. In certain embodiments, the inlet region/opening **121** can be shut via a gate valve **190** to isolate at least the first interior environment **112**. When the gate valve **190** is closed, the gases trapped in the gas segregation chamber **110** can be contained for further processing without simply escaping back through the inlet region/opening **121** and into the exterior environment **102**.

With the different gasses in the segregating gas arrangement **100**, a majority of the higher temperature condensing gasses condense on the cooling plates **105** (step **310**) preserving the lower condensing gases, such as helium, to condense in the carbon adsorber **145** (step **312**). The carbon adsorber **145** and helium can be isolated from the gas segregation chamber **110** by closing the upper gate valve **126**. In this way the carbon adsorber **145** can be removed from the adsorption gas capturing chamber **140** for further processing. Optionally, the adsorption gas capturing chamber **140** can be separated from and entirely removed from the segregating gas arrangement **100** at a junction between the gas segregation chamber **110** and the adsorption gas capturing chamber **140** (step **314**). Certain other embodiments envision a mobile gas collector (not shown) going to the segregating gas arrangement **100** and pulling the accumulated helium gas from the adsorption gas capturing chamber **140** via an access arm duct **182** in the adsorber access arm **180** that extends from the adsorption gas capturing chamber **140**. In another embodiment, the adsorption gas capturing chamber **140** can be heated to liberate the trapped helium gas in the carbon adsorber **145** when pulling the helium gas via the adsorber access arm **180**. A similar technique can be used with the gas segregation chamber **110** to collect the higher condensing temperature gases trapped therein.

Another embodiment of the present invention envisions separating multiple gas species contained within the upper chamber **140** using techniques like that of a mass spectrometer. A mass spectrometer is an apparatus for separating isotopes, molecules, and molecular fragments according to mass. A sample to be analyzed in a mass spectrometer is vaporized and ionized, and the ions are accelerated in an electric field and deflected by a magnetic field into a curved trajectory that gives a distinctive mass spectrum. Mass spectrometry (MS) is an analytical technique that is used to measure the mass-to-charge ratio of ions. In a typical MS procedure, a sample, which may be solid, liquid, or gaseous, is ionized, for example by bombarding it with a beam of electrons. This may cause some of the samples molecules to

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break up into positively charged fragments or simply become positively charged without fragmenting. These ions (fragments) are then separated according to their mass-to-charge ratio, for example by accelerating them and subjecting them to an electric or magnetic field: ions of the same mass-to-charge ratio will undergo the same amount of deflection. The ions are detected by a mechanism capable of detecting charged particles, such as an electron multiplier. Results are displayed as spectra of the signal intensity of detected ions as a function of the mass-to-charge ratio. The atoms or molecules in the sample can be identified by correlating known masses (e.g. an entire molecule) to the identified masses or through a characteristic fragmentation pattern.

Hence, certain other embodiments of the present invention envision a gas separation arrangement for separating and containing various types of gas from a collection chamber based on their different atomic masses. The chamber is connected to an ion diverter via a valve that when open permits the various types of gas to migrate from the collection chamber to an ionizer where the gas is ionized. The ionized gas is accelerated through an ion accelerator and dispensed into an ion deflector. The ion deflector comprises a magnetic field within a channel defined by a pair of split-pole magnets. The trajectory of the ions is based on the mass of the ions, which separates the ions. Multiple collectors are positioned at different locations on an exit side of the ion deflector to receive the different ions traveling along their respective trajectories.

FIGS. 5A-5E are line drawings of a gas separation arrangement connected to the adsorption gas capturing chamber. FIGS. 5A-5E are described in view of one another, hence elements described in conjunction with the gas separation arrangement 250 may be shown in one or more of the figures. The gas separation arrangement 250 generally comprises an adsorption gas capturing chamber 240 that collects gas trapped in regolith 200 and an ion diverter 251 that segregates or otherwise separates the different types of gas 265 collected in the chamber 240.

FIG. 5A is a line drawing of the gas separation arrangement connected to the adsorption gas capturing chamber consistent with embodiments of the present invention. The adsorption gas capturing chamber 240 is like the adsorption gas capturing chamber 140 of FIG. 3 but with some minor differences including a gas transmission port 252 extending from the side of the chamber 240. As shown, the chamber 240 can be connected to the lower gas segregation chamber 110 (of FIG. 1C) via the connecting passageway 125 with the option of being shut off from the gas segregation chamber 110 via the gate valve 126. The upper chamber cryogenic feed and return lines 148 are also connected to the cryogenic liquid coolant reservoir embodiment 160. The reservoir additionally feeds cryogenic fluid to segregated gas collecting receptacles (or simply "receptacles") 280, as well as all other cryogen needs of the arrangement 250. Though the adsorber access arm 180 is depicted in this embodiment, some embodiments envision no adsorber access arm 180.

The gas separation arrangement 250 comprises an ion diverter 251 connected to the chamber 240 via the gas transmission port 252. Gas 265 from the chamber 240 feeds the ion diverter 251 via a valve 254 (which in this embodiment is a needle valve but could just as easily be some other kind of valve that regulates the gas flow between the chamber 240 and the ion diverter 251). The valve 254 facilitates the movement of gas 265 from the chamber 240 into the ion diverter 251. The ion diverter 251 generally comprises the ionizer 260, the accelerator 265, a magnetic

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ion deflector 270 and receptacles 280. The ionizer 260 ionizes the gas 265 received from the chamber 240 and the accelerator 264 accelerates the ionized gas 265 to the receptacles 280 via an ion deflector 270, thus separating the gas 265 by type, which are collected in the individual receptacles 280A-280D. The receptacles 280 can be cryogenically chilled by the cryogenic fluid (from the coolant reservoir 160) circulated the feed and return lines 242 and 244. In this embodiment, there is a cryogenic fluid exchange support housing 246, that provides support for feed and return lines 242 and 244, connected to the receptacles 280.

FIG. 5B is a line drawing of a cross-section of the gas separation arrangement 250 depicting the gas transmission pathway 255 in the gas separation arrangement 250. Gas 265 from the chamber 240 is under a higher pressure than the pressure on the other side of the needle valve 254 (to the right of the needle valve 254) thereby causing gas 265 to flow to the right along the gas flow pathway 225 depicted with a thicker arrowed line. Past the needle valve 254, the gas 265 from the chamber 240 is ionized in an ion chamber 260 and then accelerated via charged plates 266 (of FIG. 5D) in an accelerator 265. The ionized gas 265 is launched into an entry location 194 (of FIG. 5C) of the channel 276 defined by a spacing 274 between a pair of split-pole magnet plates 272A and 272B, such as permanent magnets. Notably, the two magnet plates 272A and 272B generally comprise the ion deflector 270 because the magnetic field generated between the split-pole magnetic plates 272 (which are oppositely polarized) bend the trajectory of the ions based on the mass of the ions in the same way a mass-spectrometer works. Hence, the trajectory of the lighter ions bends more in the presence of the magnetic field than the heavier ions as shown by the four arrows. The trajectory of each ion type in the presence of the magnetic field is easily predicted, hence the placement of the receptacles 280 at the channel exit location 292 are appropriately placed on the distal end of the split-pole magnet pair 272A and 272B. In this embodiment, the receptacles 280 are cryogenically chilled via liquid cryogen pumped through the cryogen feed and return lines 242 and 244 by the pump 245. Cryogenically cooling the ion traps in the receptacles 280 improves trapping the ions dispensed therein.

With respect to the chamber 240, shown therein is a chamber heater 238 that when turned on liberates the gas 265 trapped in the chamber's carbon adsorber 145. In this embodiment, the liberated gas 265 is trapped in the chamber environment 142 by the gate 124 when the gate valve 126 is actuated shut. Upon opening the needle valve 254, gas 265 can be regulated to flow into the ion diverter 251 where the gas 265 from the chamber environment 142 is ionized and separated and captured in the receptacles 280A-D.

FIG. 5C is a line drawing depicting the gas separation arrangement 250 in a cross-section side view consistent with embodiments of the present invention. Briefly, the gas separation arrangement 250 depicts elements of the adsorption gas capturing chamber 240 and the elements of the ion diverter 251. In general, various gas types 265 that make it into the chamber 240 adsorb in the carbon adsorber 145 (or some other adsorber that is intended to capture or otherwise trap gas 265). The ion diverter 251 separates the type of gas 265 A-D, which can then be collected by gas type 265A-D in the receptacles 280A-D.

One embodiment envisions the needle valve 254 being closed while gases that migrate into the chamber 240 are collected in the carbon adsorber 145. Based on some triggering event (such as enough gas 265 that is believed collected in the carbon adsorber 145, or if gas 265 being

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collected from a location is believed to be depleted from the location, or after a predetermined amount of time, or based on some other external input like a person or computer program making decisions), the gate valve 126 in the connecting passageway 125 is closed, such as by a controller 5 (not shown) operating via a computer program in sync with the rest of the system 250. With the gate valve 126 and the needle valve 254 closed, the chamber heater 238 is turned on (such as by a controller), which heats up the carbon adsorber 145 thereby liberating the gas trapped therein. Accordingly, pressure from the liberated gas builds in the chamber environment 142. With an elevated pressure in the chamber environment 142, the needle valve 254 is opened allowing a controlled amount of the gas 265 to pass through the gas transmission port 252 and into the ionizer 260 by way of the needle valve 254. 10

As also shown in FIG. 5D, the ionizer 260 has an ionizer region 258 where electrons are emitted (via an electron emitter 256) to bombard the base-particles (gas 265) thus creating ionized gas. The ionized gas is accelerated via an electric field produced by accelerator plates 266 (such as magnets) in the accelerator 264 and directed through the accelerator apertures 268 located in the center of the accelerator plates 266. The accelerated ionized gas 265 A'-D' hurls towards the gap/channel 276 in between the split-pole magnets 272A and 272B. The split-pole magnets 272 redirect or otherwise bend the trajectory 275A-D of the ionized gas 265 A'-D' in the angled magnet housing 278 in predictable angle β +/-some offset, depending on the mass of the ionized gas 265 A'-D'. In this example, there are four ionized gas types 265A'-D' that each have a different mass and therefore a different trajectory 275A-D. For example, the target gases 265A-D that are segregated and collected in this embodiment may consist of H-1 (protium), H-2 (deuterium), He-3, He-4. Their respective trajectories 275A-D are drawn as trajectory H-1 275A, trajectory H-2 275B, trajectory He-3 275C, and trajectory He-4 275D, each of which lead into their respective segregation gas collecting receptacle 280A-D. In other words, the four segregated gas collecting receptacles 280A-D each capture the gas ions 265A'-D' based on ion type, which are dispensed based on the ion mass. The ion diverter 251 further comprises a tailings port 282, which can be used to capture or dispense any other heavier ionized gases into the exterior environment 102. The gas 265 from the chamber 240 travels together along a gas transmission pathway first portion 255A before diverging based on relative mass in a gas transmission pathway second portion 255B between the split-pole magnets 272. Certain embodiments envision that the internal pressure of the channel/gap 276 is essentially the same as the environment external 102 to the channel 276 because the system 250 is being used in an extraterrestrial environment, such as the Moon. 15

FIG. 5D is a higher resolution line drawing of the elements between the chamber 240 and the ion deflector 170 consistent with embodiments of the present invention. As shown, the gas transmission pathway 255 starts at the gas transmission port 252 and leads into the needle valve 254 where it goes into the ionizer 260. The ionizer 260 comprises a pair of electrical leads 262 that powers an electron emitter 256, which ionizes the gas 265 A-D entering in the ionizer chamber/region 258 from the needle valve 254. The ions 265A'-D' enter the ion accelerator 264 via an ionizer-to-accelerator port 259 where the ions 265A'-D' are accelerated through the magnetic accelerator plates 266 in the accelerator 264. The ions 265A'-D' are emitted through the accelerator aperture 268 and into gap/channel 276 between the split-pole magnets 272. 20

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FIG. 5E is a line drawing depicting a cross-section of the receptacle arrangement 280 consistent with embodiments of the present invention. Each of the segregated gas collecting receptacles 280A-D are connected (via circulation tubes) to the cryogenic feed and return lines 242 and 244, as shown in FIG. 5C. In more detail, the segregated gas collecting receptacle 280 generally comprises a receptacle adsorber 285 (which can be an activated carbon, carbon, zeolite, just to name several porous adsorbers commercially available) housed inside of a receptacle housing 286. The receptacle adsorber 285 can be maintained at cryogenic temperatures, which in certain embodiments uses a liquid helium maintained below 2° C. The liquid cryogen is circulated through the receptacle cryogen reservoir 284 in through a receptacle feed line 282A that is connected to the cryogen feed line 242 and out through a receptacle return line 282B that is connected to the cryogen return line 244. When the receptacle shutoff valve 288 is open, ionized gas is directed (from the split-pole magnets 272) into the receptacle 280 via the receptacle inlet port 290. Upon harvesting the gas 265 trapped in the receptacle adsorber 285, the shutoff valve 288 is closed. One embodiment envisions harvesting the gas 265 trapped in the receptacle 280 by removing the receptacle 280 from the gas separation arrangement 250, such as by a robot or manually. Another embodiment envisions harvesting the gas 265 trapped in the receptacle 280 by heating up the receptacle adsorber 285 via a heater, not shown, connected to or in the receptacle housing 286 and drawing the liberated gas 265 out from an access line 281. The heater causing the gas 265 trapped in the receptacle adsorber 2 25

Certain other embodiments envision a different adsorber in the receptacle 280, such as a zeolite adsorbing material, which could be used instead of activated carbon in the adsorber 145 in the chamber 240. Zeolites are microporous solids that can be used for gas separation and/or gas collection. While zeolites do occur naturally as minerals, most commercially available zeolite materials today are synthesized in a laboratory. Zeolites are aluminosilicates, comprised of aluminum, silicon, and oxygen with included hydrogen ions or metallic ions such as sodium, calcium, and magnesium. Zeolite crystals exhibit an open cage structure with openings of molecular dimensions (typically measured in Angstroms). Unlike silica gel or activated carbons that usually have irregular pores described by a broad pore size distribution, zeolites have very regular and consistent pore geometry in a very narrow size range. Often, zeolites are called "molecular sieves" because their pores will admit small molecules and bind them while excluding larger molecules that cannot fit into the pores. Zeolites are stable at high temperatures (200° C.-400° C., for example) and so can be regenerated by heating. Heating the zeolite to high temperatures will desorb the trapped gases. 30

FIGS. 6A-6E are line drawings that illustratively depict another gas separation arrangement consistent with embodiments of the present invention. FIG. 6A is a cross-section of a front view of a gas separation and collection arrangement 300 (or simply separation arrangement) comprising four gas separation units 311 connected to a gas collection system 100. As previously discussed, the upper chamber 140 comprises segregated gas 265 that was not condensed or otherwise separated out on the cooling plates 105 in the lower chamber 116. The lower chamber 116 receives gas liberated from the regolith 200 through the inlet region 121. In this arrangement 100, the upper chamber 140 of the gas collection system 100 contains four gaseous species 265A-D, which by way of example are He-3 265A, He-4 265B, Hydrogen 265C and Argon 265D. It should be appreciated 35

that the upper chamber 140 can just as easily include different species of gases and/or greater of fewer species of gases than in this example based on the temperature of the cooling plates 105 in the lower chamber 116.

As shown, the separation arrangement 300 is linked to the gas collection system 100 via a first transfer tube 330A (generically called out as 330), wherein gas 265 from the upper chamber 140 can pass into the first separation tank 301A. The transfer tubes 330 comprise an inlet port 334 at an inlet end and an outlet port 335 at an outlet end. The first transfer tube 330A provides fluid communication between the first separation tank 301A and the upper chamber 140 when the transfer tube cutoff valve 332 is open, meaning fluid (gas 265) can flow into the inlet port 334 at the upper chamber 140 and out through the outlet port 335 in the first separation tank 301A.

In this embodiment, the separation tanks 301 are housed by respective outer tank housings 302. The tank housings 302 provide structural support for support legs 340, transfer tubes 330 and their respective heat exchangers 320. The support legs 340 are one embodiment of a supporting element holding up the tank housings 302, which are envisioned to rest atop the surface of regolith 200. Other supporting elements could be a rail or box structure with a sled, wheels, tracks, or some frame that is shaped differently than legs but serves essentially the same purpose.

FIG. 6C is an angled semi-cross-section view of a gas separation unit 311 depicted to help explain the front view of a gas separation unit 311 of FIG. 6A. As shown by the flow direction arrow 336A, four species of gas 265A-D enter the separation tank 301 via the transfer tube 330 on the left side of the gas separation unit 311. Meanwhile, the transfer tube cutoff valve 332 is closed to retain the four species of gas 265A-D inside of the separation tank 301. A heat exchanger 320, which in this case is a pulse tube cryocooler commercially available by Bluefors, Corporation of Helsinki, Finland, generally comprises a compressor system 322, liquid helium reservoir 325, return line 324, feed line 326, and the adiabatic expansion pulse tube 323.

In general, pulse tube coolers are refrigeration devices that achieve cryogenic temperatures (below 123 K) without moving parts in the cold region, making them reliable and low-maintenance solutions for various applications in science and industry. Pulse tube cryocoolers operate on the principle of cyclic compression and expansion of a working gas (in this case helium) to achieve cooling. The cooling effect is generated through the thermodynamic processes of the gas as it oscillates within the system. The absence of moving parts in the cold region 321 is a distinctive feature that enhances reliability and longevity. The key principle behind their operation is the transfer of heat from the cold end 321 to the warm end 327 of the pulse tube 323 through gas dynamics and phase shifts introduced by specific components.

The compressor 322 compresses the helium gas to create oscillating pressure waves necessary for the refrigeration cycle. By generating high-pressure oscillations, the compressor initiates the thermodynamic process that drives the cooling cycle by alternately compressing and expanding the working gas (such as helium or some other gas that fulfills the purpose of target gas liquefaction). The aftercooler (not shown) is a heat exchanger positioned immediately after the compressor 322. Its role is to cool the compressed helium gas before it enters the regenerator (not shown). By reducing the gas temperature, the aftercooler enhances the efficiency of the subsequent heat exchange processes. The regenerator is a key component that significantly impacts the efficiency

of the pulse tube cryocooler 320. It often comprises a porous medium, such as metal meshes or rare-earth compounds, which has a high heat capacity, that temporarily stores heat from the gas during the compression phase and releases it during the expansion phase. During the compression phase, the regenerator temporarily stores heat from the gas, and during the expansion phase, it releases this stored heat back to the gas. This cyclical heat exchange process is fundamental to maintaining the temperature differential required for effective cooling. The pulse tube 323 is a cylindrical tube where the gas undergoes adiabatic expansion and compression of the gas take place, contributing to the cooling effect. It connects the warm end 327 and the cold end 321 of the system 320. As the gas oscillates within the pulse tube, it undergoes adiabatic processes, meaning there is no heat exchange with the surroundings during these oscillations. This leads to temperature variations that contribute directly to the cooling effect. The inertance tube or orifice is an essential component that introduces a phase shift between the pressure and the mass flow of the gas, driving the efficiency of the cooling process by controlling the timing of gas movements. The phase shift is crucial for the timing of gas movements within the pulse tube, thereby enhancing the refrigeration efficiency. The inertance tube or orifice ensures that the gas absorbs heat at the cold end and releases it at the warm end 327 in a manner that drives the cooling effect. The reservoir 325 acts as a buffer volume that stabilizes the pressure oscillations within the system 320 and maintain smooth operation of the system. It works in conjunction with the inertance tube or orifice (not shown) to maintain the necessary phase shift and pressure dynamics. Heat exchangers are placed at both the warm and cold ends 321 and 327 of the pulse tube 323. These components facilitate the transfer of heat between the working gas and the external environment 102. At the warm end 327, the heat exchanger expels heat absorbed by the gas, while at the cold end 321, the heat exchanger allows the gas to absorb heat from the environment 102 and/or the gas 265A-D being cooled. The cold heat exchanger (not shown) extracts heat from the gas 265A-D inside of the separation tank 301.

With continued reference to FIG. 6C, a liquid He-4 supply (which can be solely contained within the reservoir 325 or provided by the reservoir 160 that potentially distributes liquid He to all the gas separation units 311, for example) acts, or functions, as the heat exchange medium. In the present embodiment, a liquid He-4 supply in the liquid helium reservoir 325 is dedicated to its corresponding gas separation unit 311. In other words, the first gas separation unit 311A has its own liquid He-4 supply, which is different from a liquid He-4 supply for the second gas separation unit 311B, and so on. In this way, temperature is tailored for each of the gas separation units 311. An optional embodiment envisions sharing a common liquid He-4 supply different from the reservoir 160 that sits atop the gas collection system 100.

A target gas is separated from the other gas by selectively liquefying the target gas at a temperature above the liquefaction temperatures of the other species mixed with the target gas. In the present embodiment, the liquid He-4 supply is cycled or otherwise pumped (via a pump (not shown but could be anywhere in the liquid helium pathway) through the cooling jacket 303, which cools the separation tank interior 306 just below the gas to liquid transition temperature of the target gas but above the gas to liquid transition temperature of the other gaseous species present in the separation tank 301. The separation tank interior 306 is the volume or space inside of the separation tank 301. In this

embodiment, the cooling jacket **303** is a coiled tube that is in contact with the outside of the separation tank **301**, which facilitates heat exchange through conduction. Certain embodiments envision the cooling jacket **303** and separation tank **301** being aluminum or copper or some other metal/material that has a high flux rate, i.e., conducts well.

For example, with respect to FIG. 6A in view of FIG. 6C, all the gas species **265A-D** will be present in the first gas separation unit **311A** (the four gas species **265A-D** being received by the upper chamber **140** via the transfer tube **330**). In the forgoing example the four species of gas **265A-D** are envisioned to be He-3 **265A**, He-4 **265B** hydrogen (H) **265C** and neon (Ne) **265D**. The first gas separation unit **311A** is configured to target separating out the warmest gas species, which in this example is Ne **265D**. Ne **265D** has a gas to liquid transition temperature (liquefaction temperature) at 27° Kelvin. Therefore, by maintaining the temperature of the first separation tank interior **306A** in the range between 21°-27° Kelvin, the Ne **265D** will liquify and pool in the bottom **305** of the first separation tank **301A**. The remaining gas species **265A-265C**, which stay gas because their respective liquefaction temperatures are below 20° Kelvin are then flowed **336B** to the next gas separation unit **311B** when the next transfer tube cutoff valve **332B** is opened. The liquid Ne **265D** (which accumulates/pools in the tank bottom **305**) can be extracted from the collection port **310** via the collection conduit **308** when the gas collection valve **312** is opened. Some embodiments envision the first (inlet) transfer tube cutoff valve **332A** opened, while the second (exit) transfer tube cutoff valve **332B** is shut, to populate the first separation tank **301A** with gas **265A-D**. After waiting for a length of time that is sufficient to liquify the Ne **265D**, the first transfer tube cutoff valve **332A** can be closed and the second transfer tube cutoff valve **332B** can be opened.

Following a period of time for at least a portion of the remaining gas **265A-C** to migrate into the next separation tank **301B**, the second transfer tube cutoff valve **332B** can be closed and the first transfer tube cutoff valve **332A** reopened. This can be repeated over and over while the liquid Ne **265D** accumulates in the bottom **305** of the first separation tank **301A**. The process is likewise repeated but with a second separation tank interior **306B** being held between 5°-20° Kelvin, which will liquify H **265C**. With the H **265C** condensed/liquified out, the third transfer tube cutoff valve **332C** is opened to migrate the He-3 gas **265A** and He-4 gas **265B** into the third separation tank **301C**.

The He-3 **265A** and He-4 **265B** separation tanks **301C** and **301D** are a little different than the first two separation tanks **301A** and **301B** in that the fourth (He-4 **265B**) collection tank **301D** is in fluid communication with the third (He-3 **265A**) separation tank **301C** via a capillary transfer tube **331**. The third separation tank **301C** can be cooled below 4° Kelvin, which is the gas to liquid temperature of He-4 **265B**. The He-4 **265B** liquifies and pools in the tank bottom **305C**. As a superfluid, liquid He-4 is known to readily climb through capillary tubes or other tubes, such as a non-capillary tube (not shown), located with a tube inlet port at the tank bottom **305C**. One embodiment envisions the capillary transfer tube **331** inlet port (or regular tube inlet port) within 0.1 inches from the tank bottom **305C**, so long as the pooled liquid He-4 **265B** can contact and climb through the transfer tube into the collection tank **301D**. Some embodiments contemplate the liquid He-4 **265B** turning back into He-4 gas **265B** to flow into the collection tank **301D** where it can be reliquified to pool at the bottom **305D**

of the collection tank **301D**. The liquid He-4 **265B** can then be harvested via the collection port **310D** and the end of the collection tube **308D**.

The He-3 gas **265A** is envisioned to be collected via a collection port **310C** at the end of the collection tube **308C** when the collection cutoff valve **304C** is opened. In this embodiment, the He-3 is in a gaseous state but could just as easily be a liquid harvested from the tank bottom **305C**. Certain embodiments envision the third collection tank **301C** devoid of a cooling jacket **303C** and instead cycling the He-3 gas **265A** and He-4 gas **265B** directly through the heat exchanger **320**, as shown in FIG. 6D. In this embodiment, the heat exchanger **320** does not need a reservoir **325** because the He-3 gas **265A** and He-4 gas **265B** is cooled in the heat exchanger **320** and replenished as cold gas or liquid in the third collection tank **301C**.

In the embodiment of FIG. 6A, when the third transfer tube cutoff valve **332C** is opened, He-3 gas **265A** and He-4 gas **265B** migrate/flow into the third separation tank **301C**. The fourth transfer tube cutoff valve **332D** between the third collection tank **301C** and the fourth collection tank **301D** can either remain open or closed depending on the user's preference. If the fourth transfer tube cutoff valve **332D** is closed, He-4 gas **265B** will essentially not migrate back into the third separation tank **301C**, however it is conceived that very little He-4 gas **265B** will flow back into the third separation tank **301C** regardless of if the fourth transfer tube cutoff valve **332D** is present or not. Accordingly, another embodiment contemplates no fourth transfer tube cutoff valve **332D** between the third collection tank **301C** and the fourth collection tank **301D**.

FIG. 6B is a top view of the gas separation arrangement **300** depicting the compressors **322** of the heat exchangers **320** connected to and extending to the right of the gas collection system **100**, which prominently shows the upper and lower chamber gate valve **126** and cryogenic liquid coolant reservoir **160**. The transfer tubes **330** and their associated transfer tube cutoff valves **332** are shown connecting the gas separation units **311** in between the heat exchangers **320**. For reference, the support legs **340** and return lines **324** are shown.

FIG. 6D is an isometric view of another gas separation arrangement **350**, which is a slight variation of the gas separation arrangement **300** consistent with embodiments of the present invention. The tank housings **302** and the cooling jackets **303** are cross sectioned for better viewing. In this embodiment, the third separation tank **301C** is devoid of a cooling jacket. Rather, He-3 gas **265A** and He-4 gas **265B** in the third separation tank **301C** are cycled directly through the heat exchanger **320C** being cooled to liquefaction temperatures. In other words, the He gas **265A** and **265B** that is being harvested is also being used to chill the system **311C**, as reflected by the feed/inlet flow arrow **339** coming out of the feed line **326** and the return/outlet flow arrow **337** going into the return line **324**. The gas collection system **100** along with certain associated elements, such as the upper gas collection chamber **140**, lower gas segregation chamber **110** and the inlet region **121** are labeled for reference. The heat exchangers **320** of each gas separation unit **311** are not cross-sectioned. From this perspective, it is easily seen the interconnection and fluid communication of the gas separation units **311** via the transfer tubes **330**. It is also readily shown how the collection tubes **308** provide fluid communication between the bottom side **305** of the separation tanks **301** and the respective collection ports **310**. The third and fourth gas separation units **311C** and **311D** are for separating He-3 gas **265A** from He-4 gas **265B**. Certain embodiments contem-

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plate the number of gas separation units **311** reflecting the number of target gases being separated out. Hence, if only He-3 gas **265A** and He-4 gas **265B** are being targeted, the arrangement **300** will require only two gas separation units **311**. It should be appreciated that because He-3 gas is so rare compared with He-4 gas, decent gas separation is considered 50%. Accordingly, in certain embodiments it is a reasonable goal to enrich He-3 as opposed to separate wherein better separation can be done off-site and away from the arrangement **300**.

FIG. **6E** is an isometric view of a portion of the third separation tank **301C** and the fourth separation tank **301D** cross sectioned. As shown in this embodiment, the third separation tank **301C** has no cooling jacket. Because He-4 **265B** is a superfluid, liquified He-4 **265B** that pools at the separation tank bottom **305C** of the third separation tank **301C** will climb into the capillary tube inlet port **338** of the capillary transfer tube **331** where the He-4 **265B** can warm up enough to transition back into gas at the He-4 transfer tube **342**. The He-4 transfer tube **342** comprises a transfer tube cutoff valve **332D** that can be opened to let the He-4 gas **265B** migrate into the fourth separation tank **301D**. Certain embodiments envision no transfer tube cutoff valve **332D**. Instead, facilitating constant and free flow of the He-4 gas **265B** migrate into the fourth separation tank **301D**.

The capillary tube inlet port **338** is at a capillary tube to base separation **344** at or below the level of the pooling He-4 liquid **265B** to facilitate He-4 **265B** migration through the capillary transfer tube **331**. The capillary tube to base separation **344** can be basically/barely touching the bottom side **305C** of the third separation tank **301C**, such as 0.001 inches from the bottom side **305C** to something greater, such as 0.5 inches. Other embodiments envision the capillary tube inlet port **338** being at the inner wall **346** of the third separation tank **301C**, wherein the liquid He-4 **265B** will climb the inner wall **346** and into the capillary tube inlet port **338**. Other embodiments contemplate a coating on the inner wall **346** that prevents the liquid He-4 **265B** from climbing up the inner wall **346**. Certain embodiments envision the He-3 **265A** and He-4 **265B** coexisting as cryogenic liquid at the bottom **305C** of the third separation tank **301C**, but wherein only the liquid He-4 **265B** climbs into the capillary tube inlet port **338** due to its superfluid nature.

With a high concentration of He-3 **265A**, after a significant portion of the He-4 **265B** is transferred into the fourth separation tank **301D**, any liquified He-3 **265A** can be heated in the separation tank interior **306C** to convert back to gas, via a heater or simply halting active cooling by the heat exchanger **320**. The He-3 gas **265A** can then be harvest via the collection tube **308C** extending through the third separation tank **301C** and the third tank housing **302C**. Meanwhile, the He-4 gas **265B**, entering the fourth separation tank **301D** via the He-4 transfer tube exit port **348**, can be chilled to liquify via the cooling jacket **301D** to pool in the fourth tank bottom **305D**. The pooled liquid He-4 **265B** can then be harvested via the collection tube **308D**, shown in FIG. **6A**.

With the present description in mind, below are some examples of certain embodiments illustratively complementing some of the apparatus embodiments discussed above and presented in the figures to aid the reader. Accordingly, the elements called out below are provided by example to aid in the understanding of the present invention and should not be considered limiting. The reader will appreciate that the below elements and configurations can be

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interchangeable within the scope and spirit of the present invention. The illustrative embodiments can include elements from the figures.

In that light, certain embodiments of the present invention envision a gas separation arrangement **300** comprising a chamber **140** connected to a plurality of progressively chilled separation tanks **301** that successively separate out different species of gas **265**. The chamber **140** comprises at least He-3 gas **265A** and He-4 gas **265B**, as shown in FIGS. **5C** and **6A**. A first separation tank **301C** is linked to the chamber **140** via a first transfer tube **330C** and a second separation tank **301D** is in liquid communication with the first separation tank **301A** via a capillary transfer tube **331**, as shown in FIGS. **6E** and **6F**. Liquid communication means that liquid is in contact with the elements the liquid flows through or wherein the elements directly facilitate the movement of liquid. For example, though liquid may flow through a transfer tube **330** that passes through an outer tank housing **302**, it should be appreciated that the outer tank housing **302** is not in liquid communication with the fluid by virtue of the transfer tube **330** passing there through. A first heat exchanger **320C** comprises a first He-4 supply (for example, in the liquid helium reservoir **325** that can be maintained at a first temperature), the first He-4 supply is configured to maintain the first separation tank **301C** at a first tank temperature below a He-4 liquification temperature, as shown in FIG. **6A**. A second heat exchanger **320D** comprises a second He-4 supply (that can be maintained at a second temperature), the second He-4 supply is configured to maintain the second separation tank **301D** at a second tank temperature below the He-4 liquification temperature, as shown in FIG. **6A**. A first collection port **310C** is in fluid communication with a first interior **306C** of the first separation tank **301C**, wherein the first collection port **310C** is configured to collect the He-3 gas **265A**, as shown in FIG. **6D**. A second collection port **310D** is in fluid communication with a second interior **306D** of the second separation tank **301C**, wherein the He-4 gas **265B** configured to be collected from the second collection port **310D**, as shown in FIG. **6D**.

The gas separation arrangement **300** can further comprise at least a third separation tank **301B** that is interposed between the first separation tank and the chamber **140**, wherein the third separation tank **301B** is configured to receive the He-3 gas **265A** and the He-4 gas **265B** before the first separation tank **301C**. The third separation tank **301B** comprises a third heat exchanger **320D**, which has a third He-4 supply at a third temperature. The third He-4 supply is configured to maintain the third separation tank **301B** at a third tank temperature that is above the He-4 liquification temperature, as shown in FIG. **6A**.

The gas separation arrangement **300** further imagines the capillary transfer tube **331** comprising a capillary distal port **338** within 0.5 inch from a bottom surface **305C** of the first separation tank **301C**, as shown in FIG. **6A**.

The gas separation arrangement **300** further contemplates the first separation tank **301C** being in contact with a first cooling jacket **303C** that is in fluid communication with the first heat exchanger **320C** (meaning the liquid helium from the liquid helium reservoir **325** flows therethrough). The first heat exchanger **320C** is configured to cycle the first He-4 supply at the first temperature through the first cooling jacket **303C**, as shown in FIG. **6A**. In addition, the first heat exchanger **320C** can comprise a pump (not shown) that is configured to cycle the first He-4 supply through the first cooling jacket **303C**.

The gas separation arrangement **300** further envisions the first collection port **304C** being connected to the inner

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chamber 301C via an exhaust conduit 308C, as shown in FIG. 6A. There can further be a cutoff valve 304C in the exhaust conduit 308C that is configured to block the He-3 gas 265A and He-4 gas 265B from escaping the first separation tank 301C.

The gas separation arrangement 300 further imagining the capillary tube 331 being configured to transport the He-4 gas 265B in liquid form from the first separation tank 301C to the second separation tank 301D while leaving behind the He-3 gas 265A in the first separation tank 301C, as shown in FIG. 6D.

The gas separation arrangement 300 further pondering the first He-4 supply being configured to maintain the first separation tank 301C at a first tank temperature below a He-3 liquefaction temperature, wherein the He-3 liquefaction temperature is colder than the He-4 liquefaction temperature.

The gas separation arrangement 300 further contemplating a second collection port 310D being located at a second bottom side 305D of the second separation tank 301D, wherein the second collection port 310D being configured to transport the He-4 gas 265B in liquid form out from the second separation tank 301D, as shown in FIG. 6D.

The gas separation arrangement 300 further imagining the first separation tank 301C being contained in a first outer housing 302C.

The gas separation arrangement 300 further contemplating the first heat exchanger 320C being a pulse tube cryocooler.

Other embodiments of the present invention envision a gas separation apparatus 300 comprising a chamber 140 connected to a plurality of progressively chilled separation tanks 301 that successively separate out different species of gas 265. The apparatus can therefore comprise a first separation tank 301C linked to a chamber 140 via a first transfer tube 330C and a second separation tank 301D in liquid communication with the first separation tank 301A via a capillary transfer tube 331 having a capillary inlet port 338 within 0.5 inches of a bottom surface 305C of the first separation tank 301C, as shown in FIGS. 6E and 6F. A first heat exchanger 320C is coupled to the first separation tank 301C, wherein the first heat exchanger 320C is configured to cool the first separation tank 301C below a first temperature where (at which) He-4 liquefies (such as 4° C.). A second heat exchanger 320D is coupled to the second separation tank 301D, wherein the second heat exchanger 320D is configured to cool the second separation tank 301D below the first temperature. A first collection port 310C is in fluid communication with a first interior 306C of the first separation tank 301C, the first collection port 310C is configured to collect He-3 gas 265A, as shown in FIG. 6D. A second collection port 310D is in fluid communication with a second interior 306D of the second separation tank 301C, the second collection port 310D is configured to collect He-4 gas 265B, as shown in FIG. 6D.

The gas separation apparatus 300 further contemplating the first heat exchanger 320C being a pulse tube cryocooler.

The gas separation apparatus 300 can further comprise at least a third separation tank 301B that is interposed between the first separation tank and the chamber 140, wherein the third separation tank 301B is configured to receive the He-3 gas 265A and the He-4 gas 265B before the first separation tank 301C. The third separation tank 301B comprises a third heat exchanger 320D that comprises a third He-4 supply that can be held at a third temperature, wherein the third He-4 supply being configured to maintain the third separation tank

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301B at a third tank temperature above the He-4 liquefaction temperature, as shown in FIG. 6A.

The gas separation apparatus 300 further envisioning the first collection port 310C being connected to the inner chamber 301C via an exhaust conduit 308C, as shown in FIG. 6A.

The gas separation apparatus 300 of claim 13 further contemplating the first separation tank 301C being in contact with a first cooling jacket 303C that is in fluid communication with the first heat exchanger 320C, wherein the first heat exchanger 320C is configured to cycle the first He-4 supply at the first temperature through the first cooling jacket 303C, as shown in FIG. 6A.

Still other embodiments of the present invention envision a method to separate and obtain He-3 265A and the He-4 265B. The method comprising a step for dispensing the He-3 265A and the He-4 265B in a gas form into a first separation tank 301C from a gas collection chamber 140, as shown in FIG. 5C; a step for cycling a first He-4 supply at a first temperature through a first heat exchanger 320C that is connected to the first separation tank 301C; a step for cooling the first separation tank 301C to a first temperature that is below a liquefaction temperature of the He-4 265B via the first He-4 supply; a step for liquefying the He-4 265B in the first separation tank 301C; a step for transporting the liquefied He-4 265B into a second separation tank 301D via a capillary tube 331 that connects the first separation tank 301C to the second separation tank 301D; and a step for collecting the liquefied He-4 265B from the second separation tank 301D via a second collection port 310D.

The method can further comprise a step for collecting the He-3 265A from the first separation tank 301C via a first collection port 310D.

The method can further comprise a step for cycling first He-4 supply through a first cooling jacket 303C that is in contact with the first separation tank 301C.

Still another embodiment of the present invention contemplates a gas separation apparatus 300 comprising a chamber 140 connected to a plurality of progressively chilled separation tanks 301 that successively separate out different species of gas 265. The gas separation arrangement 300 can include a chamber 140 that comprises a gaseous first species 265A and a gaseous second species 265B, as shown in FIG. 5C. A first separation tank 301A is connected to the chamber 140 via a first transfer tube 330A and a second separation tank 301B is connected to the first separation tank 301A via a second transfer tube 330B, wherein the second separation tank 301B is linked to the chamber 140 via the first separation tank 301A, as shown in FIGS. 6A and 6D. A first cooling jacket 303A surrounds at least a portion of the first separation tank 301A and a second cooling jacket 303B surrounding at least a portion of the first separation tank 301B, as shown in FIG. 6D. A first heat exchanger 320A comprises a first liquid cryogen, such as He-4 or liquid Hydrogen, wherein the cryogen is made to cycle through the first heat exchanger 320A to maintain the first cooling jacket 303A at a first temperature low enough to convert at least a portion of the gaseous first species 265A into a liquid first species 365A, as shown in FIG. 6D. A second heat exchanger 320B comprises a second liquid cryogen or optionally the liquid He-4, that cycles through the second heat exchanger 320B to maintain the second cooling jacket 303B at a second temperature that is low enough to convert at least a portion of the gaseous second species 265B into a liquid second species 365B, as shown in FIG. 6D. A first collection port 310A located at a first bottom side 305A of the first separation tank 301A, wherein the first collection

port 310A is configured to transport the liquid first species 365A out from the first separation tank 301A, as shown in FIG. 6D. A second collection port 310B located at a second bottom side 305B of the second separation tank 301B, wherein the second collection port 310A is configured to transport the liquid second species 365A out from the second separation tank 301A, as shown in FIG. 6D.

Meanwhile, yet other embodiments envision a gas separation arrangement 250 that is generally directed to an ion diverter 251 that is used to segregate and capture the segregated gas from an adsorption gas capturing chamber (chamber) 240. The arrangement can comprise a chamber 240 containing at least a first species of gas. The chamber 240 is connected to an ion diverter 251 via a gas transmission port 252. The ion diverter 251 generally comprises a valve 254 configured to control gas communication between the ion diverter 251 and the chamber 240. The ion diverter 251 includes an ionizer 260 adjacent to the gas transmission port 252, an ion deflector 270 comprising a first magnet plate 272A separated from a second magnet plate 272B by a separation spacing 274 defining a channel 276, and an ion accelerator 264 between the ionizer 260 and the ion deflector 270. A plurality of target receptacles 280 are disposed at an exit location 292 of the channel 276, wherein each target receptacle 280 corresponds to a different target gas. The ionizer 260 is configured to ionize the first species of gas 265A and the second species of 265B received from the gas transmission port 252. The ion deflector 270 is configured to direct the ionized first species of gas 265A' in a first controlled trajectory 275A from the accelerator 264 to the first target receptacle 280A and the ionized first species of 265B' in a second controlled trajectory 275B from the ion accelerator 264 to the second target receptacle 280B.

The gas separation arrangement 250 further imagines the controlled trajectories 275A and 275B being bent or otherwise arced to segregate the two species of gas 265A and 265B.

The gas separation arrangement 250 further envisions the first species of gas 265A being He-3 and the second species of gas 265B being He-4.

The gas separation arrangement 250 further contemplates the first and the second target receptacles 280A and 280B being cooled by liquid cryogen, such as liquid helium.

The gas separation arrangement 250 further envisioning the first and the second target receptacles 280A and 280B each comprising an adsorber 285, that in certain embodiments can be either a carbon adsorber or a zeolite adsorber.

The gas separation arrangement 250 further imagining the first controlled trajectory 275A being different from the second controlled trajectory 275B to separate out the two species of gas 265A and 265B.

The gas separation arrangement 250 can further comprise multiple other species of gas that when ionized comprise other trajectories 275C and 275D that are different than the first controlled trajectory 275A and the second controlled trajectory 275B.

The target receptacles 280A and 280B in the gas separation arrangement 250 are envisioned, in certain embodiments to be removably attached to the gas separation arrangement 250.

Other variations of the gas separation arrangement 250 envision the channel 276 comprising a pressure that is within 3 milli bar from an environment external 102.

Another embodiment of the present invention envisions a gas separator 250 comprising a gas transmission port 252 that is interposed between an ionizer 260 and a chamber 240, wherein the chamber 240 is configured to contain a first

species of gas 265A and a second species of gas 265B. Connected to the chamber 240 and ionizer 260 is an ion deflector 270 that comprises a pair of magnetic plates 272, which are separated by a channel 274. There is an accelerator 264 between the ionizer 260 and the ion deflector 270 and a valve 254 configured to control gas communication between the chamber 240 and the ionizer 260. There is a first and a second target receptacle 280A and 280B disposed at a first and a second exit location 292A and 292B, respectively, of the channel 274. The ionizer 260 is configured to ionize the first species of gas 265A that is received from the gas transmission port 252 and further configured to ionize the second species of gas 265B received from the gas transmission port 252. The accelerator 264 is configured to accelerate the ionized first species of gas 265A' and the ionized second species of gas 265B' towards the ion deflector 270. The ion deflector 270 is configured to deflect the ionized first species of gas 265A' in a first curved trajectory 275A between the accelerator 264 and the first target receptacle 280A and is configured to deflect the ionized second species of gas 265B' in a second curved trajectory 275B between the accelerator 264 and the second target receptacle 280B.

The gas separator embodiment 250 is further imagined having the first target receptacle 280A being configured to capture the ionized first species of gas 265A' and the second target receptacle 280A being configured to capture the ionized second species of gas 265B'.

The gas separator embodiment 250 is further imagined having the first curved trajectory 275A being different from the second curved trajectory 275B.

The gas separator embodiment 250 is further imagined wherein the chamber 240 contains four species of gas that comprises at least two additional species of gases in addition to the first species of gas 265A and the second species of gas 265B, the four species of gas including He-3, He-4, H-1 and H-2.

The gas separator embodiment 250 further imagines the first and the second target receptacles 280A and 280B being cooled by liquid helium.

The gas separator embodiment 250 further envisions having the first and the second target receptacles 280A and 280B each comprising an adsorber 285A and 285B. Optionally, each of the adsorbers 285A and 285B can be equipped with either a carbon adsorber or a zeolite adsorber.

The gas separator embodiment 250 further envisions the first and the second target receptacles 280A and 280B being removably attached to the gas separation arrangement 250.

Yet another embodiment of the present invention envisions a gas separator device 250 comprising a gas transmission pathway 255 having a second part 255B and a first part 255A that extends from a chamber 240 through a gas transmission port 252 before reaching an ionizer 260 and ion accelerator 264, after which (the ion accelerator 264 the second part 255B is arched through a magnetic field before terminating at a target receptacle 280. The chamber 240 is configured to contain a first gas species 265A. The ionizer 260 is configured to ionize the first gas species 265A. The accelerator 264 is configured to accelerate the ionized first gas species 265A' towards the magnetic field. The magnetic field is established between a pair of separated magnetic plates 270A and 270B, wherein the magnetic field is configured to bias the ionized first gas species 265A' along the second part 255B to the target receptacle 280A, which is configured to trap the ionized first gas species 265A'.

The gas separator device embodiment 250 can further comprise a second species of gas 265B in the chamber 240. The ion diverter 251 further can comprise a second target

receptacle **280B**. The ionizer **260** is also configured to ionize the second species of gas **265B** received from the gas transmission port **252**, wherein the ionized second species of gas **265B'** is biased in a different trajectory **275B** in the ion deflector **270** than the first species of ionized gas **265A'**.

Still another embodiment of the present invention contemplates a method comprising providing a chamber **240** that is connected to an ion diverter **251** as depicted in FIGS. **5A-5C**. The method further envisions flowing a first gas species **265A** and a second gas species **265B** from the chamber **240** to the ion diverter **251** via a gas transmission port **252**, followed by ionizing the first gas species **265A** and the second gas species **265B** in an ionizer **260**, followed by accelerating the ionized first gas species **265A'** and the ionized second gas species **265B'**, via an ion accelerator **264**, into a magnetic field generated by a pair of spaced apart magnetic plates **272A** and **272B**. The ionized first gas species **265A'** is biased in a first arc **275A**, via the magnetic field, to a first target receptacle **280A** and the ionized second gas species **265B'** is biased in a second arc **275B**, via the magnetic field, to a second target receptacle **280B**. The ionized first gas species **265A'** is directed to and captured in the first target receptacle **280A** and the ionized second gas species **265B'** is directed to and captured in the second target receptacle **280B**.

Another embodiment of the present invention contemplates a gas collection system **100** that generally comprises a gas segregation chamber **110**, at least one cooling plate **105** in the gas segregation chamber **110**, an adsorption gas capturing chamber **140** connected to the gas segregation chamber **110** and a carbon adsorber **145** in the adsorption gas capturing chamber **140**. The gas segregation chamber **110** comprises a housing **115**, wherein the housing **115** is defined by housing sides **116** that extend from a top housing surface **118** to a rim **120**. A first interior environment **112** is defined within the housing **115**, wherein the first interior environment **142** is in communication with an external environment **102** through only the rim **120**. The external environment **102** defined outside of the housing **115**. At least one cooling plate **105** is in the gas segregation chamber **110**, wherein the least one cooling plate **105** comprises a passageway **122** that is configured to accommodate cryogenic fluid. An adsorption gas capturing chamber **140** is connected to the housing **115**, wherein the adsorption gas capturing chamber **140** comprises a second interior environment **142** that is in communication with the first interior environment **112** via a connecting port **132**. A carbon adsorber **145** is in the second interior environment **142**.

The gas collection system **100** further envisions the carbon adsorber **145** being a cartridge that is removable from the gas collection system **100**.

The gas collection system **100** further contemplates that the connecting port **132** leads into a connecting passageway **125** that comprises a valve **126** configured to separate the first interior environment **112** from the second interior environment **142** when closed. One embodiment contemplates the valve **126** being a gate valve. Another embodiment contemplates the carbon adsorber **145** being in communication with the gas segregation chamber **110** only when the valve **126** is open.

Another embodiment of the gas collection system **100** contemplates the adsorption gas capturing chamber **140** being removable from the housing **115**.

The gas collection system **100** can further comprise a heating element **150** that is configured to heat granular soil **200** under the rim **120** when the rim **120** rests atop the granular soil **200**. One embodiment contemplates the heat-

ing element **150** being selected from a group consisting of a laser, a radiant heater, an ultrasonic heater, or a microwave heater.

The gas collection system can further comprise a filter **130** being disposed between the at least one cooling plate **105** and the rim **120**, the filter **130** is configured to filter non-gaseous material from entering the first interior environment **112** from the external environment **102**.

The gas collection system **100** can further comprise a lower valve **190** that when closed seals the at least one cooling plate **105** from the external environment **102**.

The gas collection system **100** further imagines the carbon adsorber being an activated carbon adsorber.

The gas collection system can further comprise a heat sink **144** being in contact with the carbon adsorber **145**, wherein the heat sink **144** comprises cryogen feed and return lines **148** that are configured to cool the carbon adsorber **145**.

The gas collection system **100** can further comprise a pump **158** and a reservoir **160** that is configured to hold cryogenic liquid.

In another aspect of the present invention, some embodiments envision a gas collection arrangement **100** that generally comprises a gas segregation chamber **110**, at least one cooling plate **105** in the gas segregation chamber **110**, an adsorption gas capturing chamber **140** and a carbon adsorber **145**. The gas segregation chamber **110** defines a first interior environment **112** when a rim **120** of the gas segregation chamber **110** rests atop regolith **200**. The gas segregation chamber **110** also comprises at least one cooling plate **105** that is configured to capture higher temperature condensing gas but not low temperature condensing gas, wherein the gases are released from the regolith **200**. The adsorption gas capturing chamber **140** defines a second interior environment **142** that is in communication with the first interior environment **112** via a connecting port **132**. The carbon adsorber **145** is in the second interior environment **142**, wherein the carbon adsorber is configured to capture the low temperature condensing gas.

The gas collection arrangement **100** can further comprise a tank **160** that is configured to contain cryogenic liquid to cool the at least one cooling plate **105** to first temperature and the carbon adsorber **145** to a second temperature that is lower than the first temperature.

The gas collection arrangement **100** further envisioning the low temperature condensing gas being helium and the high temperature gas including hydrogen and oxygen.

The gas collection arrangement **100** can further comprise a heat sink **144** that is connected to the carbon adsorber **145**, the heat sink **144** configured to cool the carbon adsorber at or below a temperature at which the low temperature condensing gas condenses.

The gas collection arrangement **100** can further comprise a valve **126** that is between the gas segregation chamber **110** and the adsorption gas capturing chamber **140**, the valve **126** is configured to cut off the communication between the first interior environment **112** and the second interior environment **142**.

Another embodiment of the present invention contemplates a segregating gas arrangement **100** generally comprising a gas segregation chamber **110**, at least one cooling plate **105** in the gas segregation chamber **110**, at least one cooling plate **105** in the gas segregation chamber **110**, and a carbon adsorber **145**. The gas segregation chamber **110** comprises a rim **120** that when resting atop regolith **200** defines a first interior environment **112**. The at least one cooling plate **105** is in the gas segregation chamber **110**, wherein the least one cooling plate **105** is maintained at a

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first temperature above 5° K, which is a condensation temperature at which higher temperature condensing gases condense. The adsorption gas capturing chamber 140 defines a second interior environment 142 that is in communication with the first interior environment 112. The carbon adsorber 145 is in the second interior environment 142 and is maintained at a second temperature below 3° K. The carbon adsorber is configured to capture the low temperature condensing gas.

The segregating gas arrangement 100 further contemplates the gas segregation chamber 110 being configured to filter out a majority of the higher temperature condensing gases from entering the adsorption gas capturing chamber 140.

Certain other embodiments of the present invention envision a gas segregating method comprising providing a segregating gas arrangement 100 comprising an adsorption gas capturing chamber 140 that is connected to a gas segregation chamber 110. The gas segregation chamber 110 comprises a housing 115 that is defined by housing sides 116 that extend from a top housing surface 118 to a rim 120. The method further comprises resting the rim 120 atop regolith 200, wherein a first interior environment 112 is defined within the housing when the rim 120 is resting atop the regolith 200. A first temperature above 5° K is maintained in at least one cooling surface 105 that is disposed in the gas segregation chamber 110 and a second temperature below 3° K is maintained at the carbon adsorber 145 in the second interior environment 142. A majority of higher temperature condensing gases are condensed in the first interior environment 112 but not a lower temperature condensing gas is not condensed in the first interior environment 112. The lower temperature condensing gas is captured in a carbon adsorber 145 that is located in the adsorption gas capturing chamber 140, wherein the lower temperature condensing gas migrates from the first interior environment 112 to a second interior environment 142 that is defined within the adsorption gas capturing chamber 140.

The gas segregating method can further comprise a step for removing the adsorption gas capturing chamber 140 from the gas segregation chamber 110.

The gas segregating method can further comprise a step for circulating cryogenic fluid through the at least one cooling surface 105 and through a heat sink 144 that is in contact with the carbon adsorber 145, the cryogenic fluid is held in a reservoir 160 of the segregating gas arrangement 100.

The gas segregating method can further comprise a step for removing the carbon adsorber 145 from the segregating gas arrangement 100.

The gas segregating method can further comprise a step for liberating the higher temperature condensing gases and lower temperature condensing gas from the regolith 200 by heating the regolith with a heater 150 that is cooperating with the segregating gas arrangement 100.

The gas segregating method further envisions the at least one cooling surface 105 being a cooling plate.

It is to be understood that even though numerous characteristics and advantages of various embodiments of the present invention have been set forth in the foregoing description, together with the details of the structure and function of various embodiments of the invention, this disclosure is illustrative only, and changes may be made in detail, especially in matters of structure and arrangement of parts within the principles of the present invention to the full extent indicated by the broad general meaning of the terms in which the appended embodiments are expressed. For

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example, the orientation of the elements and the plate can include other geometries not explicitly shown in the embodiments above while maintaining essentially the same functionality without departing from the scope and spirit of the present invention. Likewise, the materials and construction of the cooling surfaces 105 and heat sink 145 can be different but serve the same purpose without departing from the scope and spirit of the present invention. It should further be appreciated that the valves do not need to be gate valves but could be other valve construction including more than one valve, the basic construction is well known in the art and modification to present embodiments discussed can be made once a skilled artisan is in possession of the concepts disclosed herein. Moreover, the electronics and computing that enable the functionality of the gas collection system 100 are not described in detail because the electronics and computing elements either exist or are easily constructed by those skilled in the art. Moreover, in regards to the gas separation arrangement 300, though helium is as a refrigerant, other refrigerants can be used, such as hydrogen, nitrogen, etc., without departing from the scope and spirit of the present invention.

It will be clear that the present invention is well adapted to attain the ends and advantages mentioned as well as those inherent therein. While presently preferred embodiments have been described for purposes of this disclosure, numerous changes may be made which readily suggest themselves to those skilled in the art and which are encompassed in the spirit of the invention disclosed and as defined in the appended claims.

What is claimed is:

1. A gas separation arrangement comprising:

- a chamber comprising He-3 gas and He-4 gas;
- a first separation tank linked to the chamber via a first transfer tube and a second separation tank in liquid communication with the first separation tank via a capillary transfer tube;
- a first heat exchanger comprising a first He-4 supply configured to maintain the first separation tank at a first tank temperature below a He-4 liquefaction temperature;
- a second heat exchanger comprising a second He-4 supply configured to maintain the second separation tank at a second tank temperature below the He-4 liquefaction temperature;
- a first collection port in fluid communication with a first interior of the first separation tank, the first collection port configured to collect the He-3 gas; and
- a second collection port in fluid communication with a second interior of the second separation tank, the second collection port configured to collect the He-4 gas.

2. The gas separation arrangement of claim 1 further comprising at least a third separation tank interposed between the first separation tank and the chamber, wherein the third separation tank configured to receive the He-3 gas and the He-4 gas before the first separation tank, the third separation tank comprising a third heat exchanger comprising a third He-4 supply at a third temperature, the third He-4 supply configured to maintain the third separation tank at a third tank temperature above the He-4 liquefaction temperature.

3. The gas separation arrangement of claim 1, wherein the capillary transfer tube comprises a capillary distal port within 0.5 inch from a bottom surface of the first separation tank.

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4. The gas separation arrangement of claim 1, wherein the first separation tank is in contact with a first cooling jacket that is in fluid communication with the first heat exchanger, wherein the first heat exchanger configured to cycle the first He-4 supply at the first tank temperature through the first cooling jacket.

5. The gas separation arrangement of claim 4, wherein the first heat exchanger comprises a pump configured to cycle the first He-4 supply through the first cooling jacket.

6. The gas separation arrangement of claim 1, wherein the first collection port is connected to the first separation tank via an exhaust conduit.

7. The gas separation arrangement of claim 6, further comprising a cutoff valve in the exhaust conduit configured to block the He-3 gas and He-4 gas from escaping the first separation tank.

8. The gas separation arrangement of claim 1, wherein the capillary transfer tube is configured to transport the He-4 gas in liquid form from the first separation tank to the second separation tank while leaving behind the He-3 gas in the first separation tank.

9. The gas separation arrangement of claim 1, wherein the first He-4 supply configured to maintain the first separation tank at a first tank temperature below a He-3 liquification temperature, the He-3 liquification temperature colder than the He-4 liquification temperature.

10. The gas separation arrangement of claim 1, wherein a second collection port is located at a second bottom side of the second separation tank, the second collection port configured to transport the He-4 gas in liquid form out from the second separation tank.

11. The gas separation arrangement of claim 1, wherein the first separation tank is contained in a first outer housing.

12. The gas separation arrangement of claim 1, wherein the first heat exchanger is a pulse tube cryocooler.

13. A gas separation apparatus comprising:

a first separation tank linked to a chamber via a first transfer tube;

a second separation tank in liquid communication with the first separation tank via a capillary transfer tube having a capillary inlet port within 0.5 inches of a bottom surface of the first separation tank;

a first heat exchanger coupled to the first separation tank, wherein the first heat exchanger configured to cool the first separation tank below a first temperature where He-4 liquifies;

a second heat exchanger coupled to the second separation tank;

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a first collection port in fluid communication with a first interior of the first separation tank, the first collection port configured to collect He-3 gas; and

a second collection port in fluid communication with a second interior of the second separation tank, the second collection port is configured to collect He-4 gas.

14. The gas separation apparatus of claim 13, wherein the first heat exchanger is a pulse tube cryocooler.

15. The gas separation apparatus of claim 13 further comprising at least a third separation tank interposed between the first separation tank and the chamber, wherein the third separation tank configured to receive the He-3 gas and the He-4 gas before the first separation tank, the third separation tank comprising a third heat exchanger connected to a third He-4 supply configured to maintain the third separation tank at a third tank temperature above a liquification temperature of the He-4 gas.

16. The gas separation apparatus of claim 13, wherein the first collection port is connected to the first separation tank via an exhaust conduit.

17. The gas separation apparatus of claim 13, wherein the first separation tank is in contact with a first cooling jacket that is in fluid communication with the first heat exchanger, wherein the first heat exchanger configured to cycle a first He-4 supply at the first temperature through the first cooling jacket.

18. A method to separate and obtain He-3 and the He-4, the method comprising:

dispensing the He-3 and the He-4 in a gas form into a first separation tank from a gas collection chamber;

cycling a first He-4 supply at a first temperature through a first heat exchanger connected to the first separation tank;

cooling the first separation tank to a first temperature below a liquification temperature of the He-4 via the first He-4 supply;

liquifying the He-4 in the first separation tank;

transporting the liquified He-4 into a second separation tank via a capillary tube connecting the first separation tank to the second separation tank; and

collecting the liquified He-4 from the second separation tank via a second collection port.

19. The method of claim 18 further comprising collecting the He-3 from the first separation tank via a first collection port.

20. The method of claim 18 further comprising cycling first He-4 supply through a first cooling jacket in contact with the first separation tank.

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